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Defending the beauty of the Invariance Principle

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Customary stability analysis methods for nonlinear nonautonomous systems seem to require a strict condition of uniform continuity. Although extensions of LaSalle's Invariance Principle to nonautonomous systems that mitigate this condition have been available for a long time, they have remained surprisingly unknown or open to misinterpretations. The large scope of the Principle might have misled the prospective users and its application to Control problems has been received with amazing yet clear uneasiness. Counterexamples have been used in order to claim that the Invariance Principle cannot be applied to nonlinear nonautonomous systems. Because the original formulation of the Invariance Principle still imposes conditions that are not necessarily needed, this paper presents a new Invariance Principle that further mitigates previous conditions and thus further expands the scope of stability analysis. A brief comparative review of various alternatives to stability analysis of nonautonomous nonlinear systems and their implications is also presented in order to illustrate that thorough analysis of same examples may actually confirm the efficiency of the Invariance Principle approach when dealing with stability of nonautonomous nonlinear systems problems that may look difficult or even unsolvable otherwise.

Keywords: Invariance Principle; autonomous and nonautonomous systems

1. Introduction

Consider the nonlinear nonautonomous system:

$$\dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x}, t). \quad (1)$$

Following Lyapunov stability theory, one chooses a uniformly positive definite (and, of course, differentiable) function $V(\mathbf{x}(t))$, that for convenience we may simply denote by $V(\mathbf{x})$. We emphasise that $V(\mathbf{x} \neq 0) > 0$ and $V(\mathbf{x} = 0) = 0$. One then computes its derivative along the trajectories of the system (1). If the derivative $\dot{V}(\mathbf{x}, t)$ is negative definite, one concludes that $V(\mathbf{x})$ can only decrease and therefore, because $V(\mathbf{x})$ is bounded from below, all trajectories must ultimately reach the origin $\mathbf{x} = 0$. Note that, although one can choose Lyapunov functions that are not explicit functions of time, their derivative is denoted $\dot{V}(\mathbf{x}, t)$ because usually $\dot{V}(\mathbf{x}, t) = \frac{dV(\mathbf{x}(t))}{dt} = \frac{\partial V(\mathbf{x}(t))}{\partial \mathbf{x}} \frac{d\mathbf{x}(t)}{dt} = \frac{\partial V(\mathbf{x}(t))}{\partial \mathbf{x}} \mathbf{f}(\mathbf{x}, t)$ is explicit function of t .

Although Lyapunov stability approach is the customary basis of any modern stability analysis, its direct application requires fitting to the system a positive definite function whose derivative along all trajectories of the system is negative definite. If this is the case, it is easy to see that both the Lyapunov function and its derivative ultimately vanish and therefore, one can reach the desired conclusion of asymptotic stability. However, because in most nontrivial problems the derivative is at most negative semi-definite,

various extensions to the basic Lyapunov stability theory have been sought. Because early extensions of Lyapunov stability theory were only covering autonomous systems of the form $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$ (LaSalle, 1968; LaSalle & Lefschetz, 1961), various alternatives were sought for nonautonomous systems of the form $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, t)$. In particular, a successful alternative was provided by Barbalat's Lemma that under some conditions allowed concluding that the Lyapunov derivative ultimately vanishes and therefore, as many examples illustrated, asymptotic stability conclusions could be drawn. Texts on nonlinear control such as Slotine and Li (1991), Sastri (1999), Khalil (2002), Vidyasagar (2002) and others gave various formulations intending to mitigate prior conditions that would facilitate application of Barbalat's Lemma to stability analysis.

However, as this paper attempts to show, Barbalat's Lemma that deals with function theory and *not necessarily* with stability theory imposes some conditions of continuity of functions and even continuity of derivatives, that again could limit its applicability.

As recent publications illustrate, although extensions of LaSalle's Invariance Principle to nonautonomous systems have been available at least since 1976 (Artstein, 1976, 1977, 1978; LaSalle, 1976a, 1976b), they have remained surprisingly unknown for large circles of the nonlinear control community. Moreover, even if assumably known, misinterpretations of its larger mathematical scope (that covers much more than mere asymptotic stability) may

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have misled the prospective users with respect to its usefulness. It is hoped that a new extended Invariance Principle along with a comparative parallel presentation of various alternatives to stability analysis may help showing the extreme efficiency of the Invariance Principle to nonlinear systems stability analysis.

Section 2 reviews Barbalat's Lemma and LaSalle's Invariance Principle with their advantages and limitations and then presents a new Invariance Principle that extends the scope of stability analysis to larger classes of nonautonomous nonlinear systems. A Modified Invariance Principle is also presented for those cases when transient terms affect the definiteness of the Lyapunov derivative. Section 3 then attempts to extend the stability analysis to systems that are not necessarily globally asymptotically stable. Section 4 then clarifies the definitions of stability in the sense of Lyapunov versus other stability situations and Section 5 presents some summarising conclusions. Examples are provided all throughout the paper to illustrate the differences in interpretation and sharpness of various results that the different analysis approaches provide for the same systems.

2. On Barbalat's Lemma, LaSalle's Invariance Principle and new developments in stability analysis of nonlinear nonautonomous systems

2.1 Basic assumptions of this paper

As mentioned, the main goal of this paper is extension of Lyapunov results to those systems where the Lyapunov derivative is not necessarily negative definite. Therefore, it is useful to emphasise that the fundamental assumption of this paper is:

Fundamental assumption: Consider the nonlinear and nonautonomous system (1). Unless specifically mentioned otherwise, all throughout this paper we will assume that there exists a positive definite Lyapunov function $V(\mathbf{x})$ and that its derivative $\dot{V}(\mathbf{x}, t)$ along the trajectories of Equation (1) is negative semi-definite, i.e., satisfies $\dot{V}(\mathbf{x}, t) \leq 0$.

The following sections will examine what other conditions must be satisfied along the trajectories of nonlinear and nonautonomous systems in order to allow further analysis and sharper conclusions related to stability of those systems.

To this end, it could be useful to consider what kinds of limit points and limit sets one may encounter in nonautonomous systems.

For convenience, we first consider the general autonomous system $\dot{x}(t) = f(x)$. Here, any solution x_0 of $f(x_0) = 0$ makes $\dot{x}(t) = 0$ for all t , so x_0 is an equilibrium point. If a trajectory starts at x_0 or reaches it at any time, it remains there.

However, in nonautonomous systems, one must be careful, because the time-dependent terms may define various kinds of limit sets.

Consider the scalar system $\dot{x}(t) = -x^3(1 + \sin t)$. Clearly, $x = 0$ makes $\dot{x}(t) = 0$ for all t , so $x = 0$ is an equilibrium point. If a trajectory starts at $x = 0$ or reaches it at any time, it remains there, and so, here the explicit time-dependent component does not qualitatively affect the general behaviour of this system.

However, now consider $\dot{x}(t) = -x^3 e^{-t}$. Clearly, $x = 0$ makes $\dot{x}(t) = 0$ for all t , so $x = 0$ is an equilibrium point and it is very tempting and easy to reach the conclusion that this is the only limit set. However, nonautonomous systems deserve a second look and here it shows that the conclusion above is true only for finite times. As time tends to infinity, any finite x ultimately makes $\dot{x}(t) = 0$. Still, as one can see and any simulation shows, if any trajectory starts at any $x \neq 0$, the trajectory ultimately ends at another $x \neq 0$ (as, clearly, the initial velocity is not zero). Therefore, $x = 0$ is the only 'regular' equilibrium point in the 'classical' sense, although it only 'attracts' the trivial trajectory $x(t) = 0$. On the other hand, any $x \neq 0$ is a member of the 'limit set,' as it is a potentially attractive limit point for some trajectory, yet it is not an equilibrium point in the 'classical' sense and maybe such a limit point should be called 'ultimate' equilibrium point, because no trajectory that starts there or reaches the point at any finite time remains there.

Consider now $\dot{x}(t) = -x^3 + e^{-t}$. Here, $x = 0$ makes $\dot{x}(t) = e^{-t}$ for all t , so $x = 0$ is not an equilibrium point in the strict sense, as any trajectory that starts or reaches $x = 0$ at any finite time t would keep moving away from it. However, $x = 0$ is the only limit set, because as time tends to infinity, this is the only point that ultimately makes $\dot{x}(t) = 0$ and therefore, ultimately all trajectories come to rest at $x = 0$.

Therefore, in general, the aim of stability analysis is to find the *limit sets* of trajectories that are located on the trajectories.

2.2 Barbalat's Lemma

Because, as we mentioned, in most nontrivial problems the derivative is at most negative semi-definite, various extensions to the basic Lyapunov stability theory have been sought. One of these extensions was provided by Barbalat's Lemma that we state here as it appears in Slotine and Li (1991).

Lemma 2.1 (Barbalat's Lemma): *If the differentiable function $V(t)$ has a finite limit as $t \rightarrow \infty$ and if $\dot{V}(t)$ is uniformly continuous (or $\ddot{V}(t)$ is bounded), then $\dot{V}(t) \rightarrow 0$ as $t \rightarrow \infty$.*

Barbalat's Lemma is very simple and thus, very attractive. Furthermore, under some conditions, it allowed to finally show that the function $\dot{V}(t)$ ultimately vanishes and in many cases even it allowed reaching the desirable asymptotic stability or asymptotically perfect tracking conclusion. However, because Barbalat's Lemma deals with functions

and not necessarily with *systems*, it imposes conditions on continuity of functions and even of their derivative. These conditions *might* happen to hold in various systems, yet even if under less than ideal conditions they are not satisfied, it is not necessarily a result of some lack of stability.

On the other hand, as already mentioned, extensions of LaSalle's Invariance Principle to nonautonomous systems have been available at least since 1976 (see Artstein, 1976, 1977, 1978; LaSalle, 1976a, 1976b or their summary in Kaufman, Barkana, & Sobel, 1998), and have been immediately adopted and used since 1980 for such special problems as adaptive control of large space structures and other applications (Balas, 2001; Barkana, 1983; Barkana, Kaufman, & Balas, 1983; Bayard, Ih, & Wang, 1987; Ih, Wang, & Leondes, 1985; Ih, Bayard, & Wang, 1986; Ih, Wang, & Leondes, 1987; Mareels & Polderman, 1996; Mareels, Gils, Polderman, & Ilchmann, 1994). Nonetheless, as even recent publications seem to show, either the Principle has remained largely unknown, or its larger mathematical scope that covers much more than mere asymptotic stability might have misled the prospective user with respect to its usefulness. Indeed, as many control publications (including the above-cited textbooks) show, its application to control problems has been received with clear uneasiness.

2.3 LaSalle's Invariance Principle

First, it is important to emphasise LaSalle's simple and ingenious idea. Instead of dealing with the resulting Lyapunov derivative as a *function*, LaSalle's Invariance Principle goes back to establish some (amazingly) mild conditions on the *system*. Using the notation $\|\mathbf{f}\| = \sqrt{f_1^2 + f_2^2 + \dots + f_n^2}$, satisfaction of one of the following two assumptions must be checked *along the trajectories of the system* under investigation:

- (1) $\|\mathbf{f}(\mathbf{x}(t), t)\|$ is bounded for any bounded $\mathbf{x}$.
- or
- (2) $|\int_{\alpha}^{\beta} \mathbf{f}(\mathbf{x}(\tau), \tau) d\tau| = \mu(\beta - \alpha)$.

In LaSalle's formulation, the function $\mu(\tau)$ is a 'modulus of continuity,' from which one can imply that the trajectory is supposed to be a *continuous* function of time.

Note: The detailed explanation on the eventual satisfaction of the assumptions will come later, yet even at this stage it is important to refer to these conditions that seem very difficult to accept. As a pretty vast literature seems to show, even now, many researchers may still claim that 'LaSalle's Invariance Principle only holds for time-invariant systems,' while some publications point to apparent difficulties related to its eventual applications in nonautonomous systems. We will later explain why and when these basic assumptions are satisfied, yet here we only want to emphasise

the genius of LaSalle's idea that managed to continue in the spirit of Lyapunov's idea. Indeed, there is no reason for a positive definite (Lyapunov) *function* such as $V(x) = x^2$ to have a negative definite derivative *unless* one relates the function $V(x) = x^2$ to a specific system and specifies 'derivative along trajectories of the system.' Similarly, there is no reason for the function $\mathbf{f}(\mathbf{x}, t)$ to satisfy any one of the assumptions *unless* one again relates the function $\mathbf{f}(\mathbf{x}, t)$ to a system and specifies that the satisfaction of these assumptions must only be checked '*along the trajectories of the system*.' Assumption 1 is easy to check while assumption 2 is needed in case assumption 1 is not satisfied. For example, it is easy to see that a function of the form $f(x(t), t) = -x^3 \sin t$ satisfies assumption 1, while $f(x(t), t) = -x^3 e^t$ and $f(x(t), t) = -[x^3/(1-x)] \sin t$ do not. If the last two examples then do or do not satisfy assumption 2 depends on the specific system.

Under these assumptions, we can present LaSalle's Invariance Principle.

Theorem 2.2 (LaSalle's Invariance Principle): *Consider the nonlinear nonautonomous system (1). Assume that there exists a positive definite Lyapunov function $V(\mathbf{x})$ and that its derivative $\dot{V}(\mathbf{x}, t)$ along the trajectories of Equation (1) is negative semi-definite, and in particular satisfies a relation of the form $\dot{V}(\mathbf{x}, t) \leq -W(\mathbf{x}) \leq 0$. For any initial condition $\mathbf{x}(t=0) = \mathbf{x}_0$ define the domain $\Omega_0 = \{\mathbf{x} | V(\mathbf{x}) \leq V(\mathbf{x}_0)\}$. Because the Lyapunov derivative is negative semi-definite, it is clear that all system trajectories are bounded and contained within the domain Ω_0 . Now, define the domain $\Omega = \{\mathbf{x} | W(\mathbf{x}) = 0\}$. Then, if one of the assumptions 1 or 2 holds, the entire state vector $\mathbf{x}(t)$ ultimately reaches the domain $\Omega_f = \Omega_0 \cap \Omega$ (see Kaufman et al., 1998; LaSalle, 1976b).*

We note that an almost similar theorem can be formulated and proved using Barbalat's Lemma (Khalil, 2002).

2.4 The new Invariance Principle

As recently observed and as this paper will show, even the milder conditions that LaSalle's original formulation still seems to impose are difficult to satisfy in realistic applications. In particular, the apparent need for *uniform* continuity may lead the reader to the conclusion that stability of nonlinear systems is threatened by the presence of any step, square-wave or any other discontinuous input command or disturbance. It is the scope of this paper to show that many of these *apparent* conditions are overly restrictive and that in fact they are *not* necessarily needed. Also, because, as mentioned above, the scope of the Invariance Principle is very large and its conclusion in the general nonlinear systems is *not necessarily* asymptotic stability, for reader's convenience we present a new Invariance Principle for nonautonomous systems in a form that is appropriate for the problems we discuss and that further relaxes even the milder conditions of LaSalle.

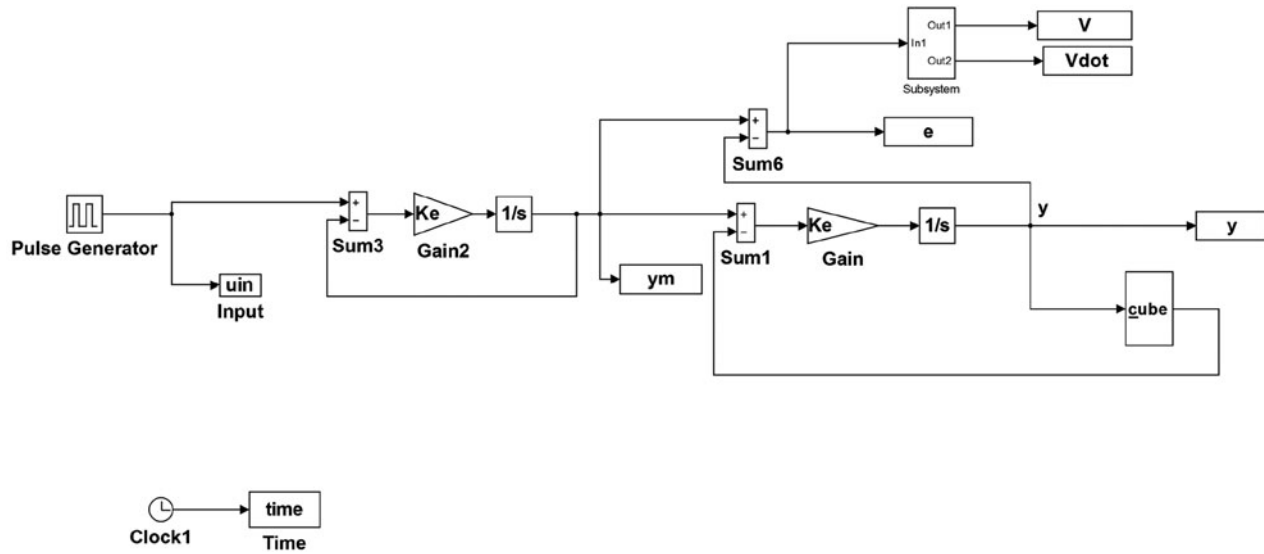


Figure 1. Simple example – continuous behaviour.

Two simple examples will be used to illustrate the following developments. First example (Figure 1) is the ‘decent,’ perfectly continuous plant

$$\dot{x}(t) = -x^3 + u(t) \quad (2)$$

$$y(t) = x(t). \quad (3)$$

The Plant is tracking the simple first-order model

$$\dot{x}_m(t) = -x_m + u_m(t) \quad (4)$$

$$y_m(t) = x_m(t) \quad (5)$$

such that $u(t) = y_m(t)$. Figure 2 shows the tracking error $e(t)$.

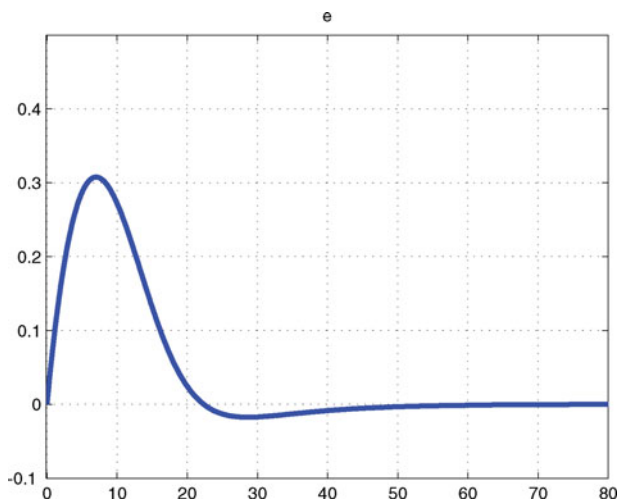
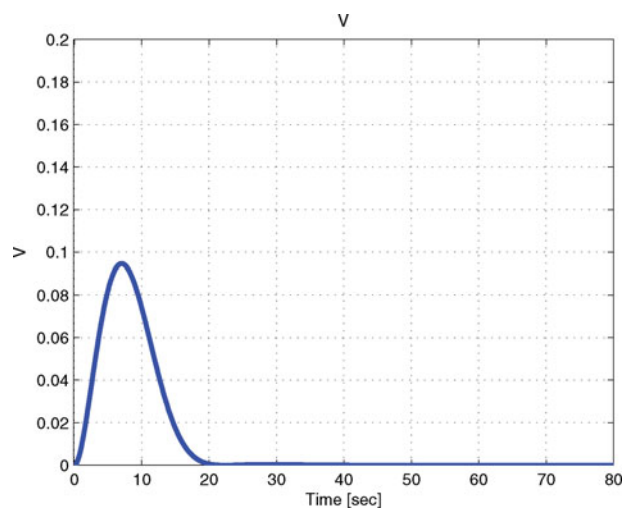


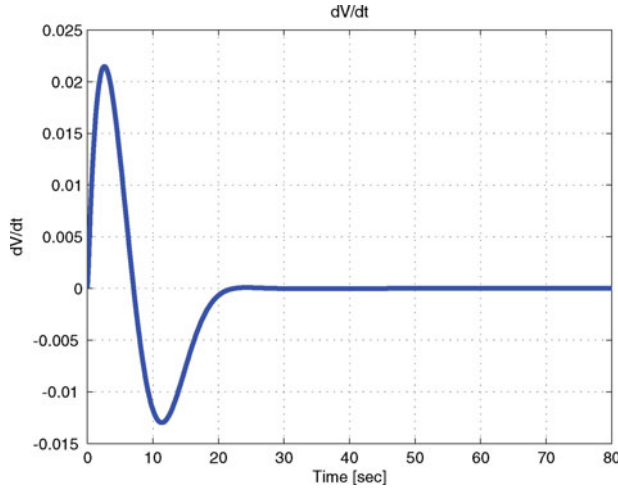
Figure 2. Simple example – tracking error.

Here, instead of writing and analysing the differential equation of the error, we compute and plot the Lyapunov function $V(e(t)) = e^2(t)$ (Figure 3) and its derivative $\dot{V}(t)$ (Figure 4). It can be seen that, after an initial transient, the plant output starts following the model output. Correspondingly, the tracking error ultimately vanishes and so do $V(e(t)) = e^2(t)$ and its derivative $\dot{V}(t)$.

In order to check how important the uniform continuity requirement is for stability, we again use the same simple example, only this time with the addition of a zero-order-hold (ZOH) processor on model output $y_m(t)$ and on plant state $x(t)$, thus making it into a sequence of successive steps function. Thus, instead of $y_m(t)$ we supply $y_{m_{zoh}}(t)$ to the plant and the plant feedback signal $x^3(t)$ is replaced by x_{zoh}^3 (Figure 5).

Figure 6 shows the tracking error $e_{zoh}(t)$.

Figure 3. Simple example – $V(x(t))$.

Figure 4. Simple example – $\dot{V}(x, t)$.

It can be seen that, after an initial transient, the plant output starts following the model output. Here, instead of writing the differential equation of the error and analysing it, we compute the Lyapunov function $V_{zoh}(e_{zoh}(t)) = e_{zoh}^2(t)$. Both the tracking error e_{zoh} and the Lyapunov function V_{zoh} (Figure 7) show a similar behaviour to the corresponding continuous system: after some transient, the discontinuous tracking error behaves like a staircase with ever diminishing steps that ultimately vanishes, even though the new Lyapunov function is only piece-wise continuous and its derivative is only a sequence of impulse functions (Figure 8). Here, the diminishing amplitudes are the computer representation of derivatives at ever diminishing steps.

This is only an illustration to motivate the following discussion and analysis. Because we deal with differential equations as representations and approximations of real-world systems, the very basic concept of continuity versus discontinuity must be discussed. Although there is no infinite velocity in real world, we take discontinuity as a proper approximation for very fast changes that cannot get any better representation. Saying this, if we treat as real the

fact that a trajectory can change its position in no time, then we must also accept as real the fact that there is also an infinite momentary speed that allowed this to happen. In other words, the δ -function, the derivative at the point of discontinuity, either we call it δ -function, δ -distribution or generalised function, must be considered as real as the discontinuity itself. The need for generalised derivatives is not new and for a more fundamental theoretical treatment of the so-called Dini derivatives and generalised derivative, see for example Bruckner (1994). For the needs of this stability analysis, we can assume that trajectories are piece-wise continuous and differentiable in the classical sense, yet we also add that at points of finite discontinuities, if $f(t = \tau_-) = a$ and $f(t = \tau_+) = b$, then $\frac{df(t=\tau)}{dt} = (b - a)\delta(t - \tau)$ and in this sense, such bounded functions are differentiable everywhere.

With this in mind, we do not necessarily look for continuous solutions of the differential equations, but rather for all possible solutions, be they continuous, discontinuous, etc. Thus, $\dot{x}(t) = -x^3 + 2\delta(t) + 3\delta(t - 3)$ has solutions even though $f(x, t) = -x^3 + 2\delta(t) + 3\delta(t - 3)$ is neither continuous nor bounded. In order to integrate this equation, one must recall that the presence of $2\delta(t)$ implies that at time $t = 0$, the system trajectory is suddenly switched from the initial value $x(0_-)$ to the new value $x(0_+) = x(0_-) + 2$ from where the integral is just $\dot{x}(t) = -x^3$ that continues up to $t = 3$, when $x(t)$ is suddenly switched from the value $x(3_-)$ to the new value $x(3_+) = x(3_-) + 3$, from where the integral again is just $\dot{x}(t) = -x^3$ (Figure 9).

Another point to notice is the way trajectories may reach an equilibrium point. In autonomous systems, if a trajectory does not start at an equilibrium point, it reaches it only as time tends to infinity. However, because of the eventual existence of discontinuous trajectories, it is useful to notice that trajectories may reach the equilibrium point at any finite time, as the illustration of $\dot{x}(t) = -x^3 + 2\delta(t) - 0.4848\delta(t - 2)$ illustrates (Figure 10).

Because this point could be controversial and, therefore, difficult to digest and accept, and because the aim

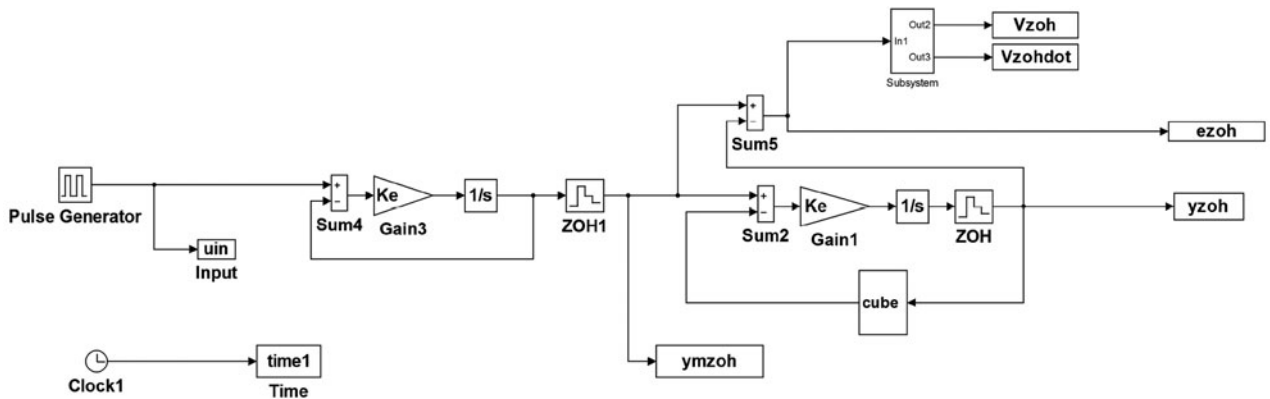


Figure 5. Simple example – discontinuous behaviour.

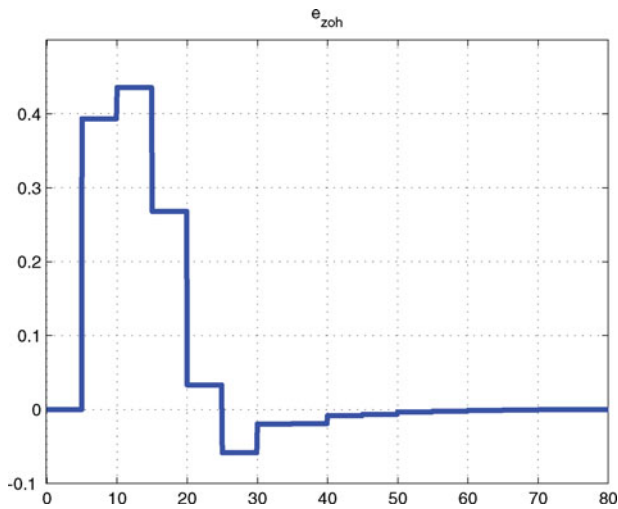
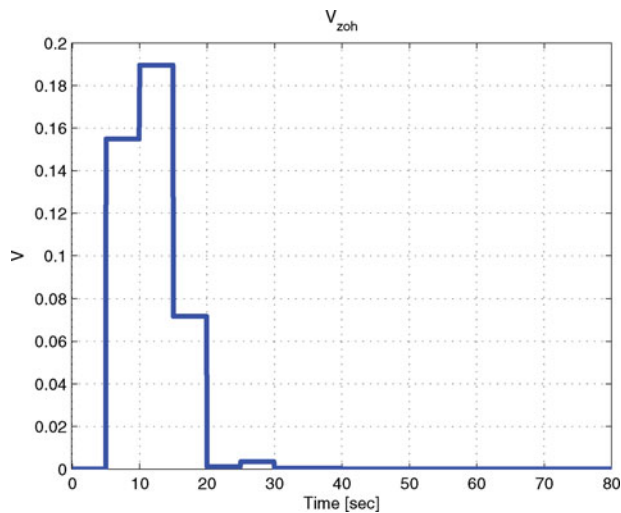
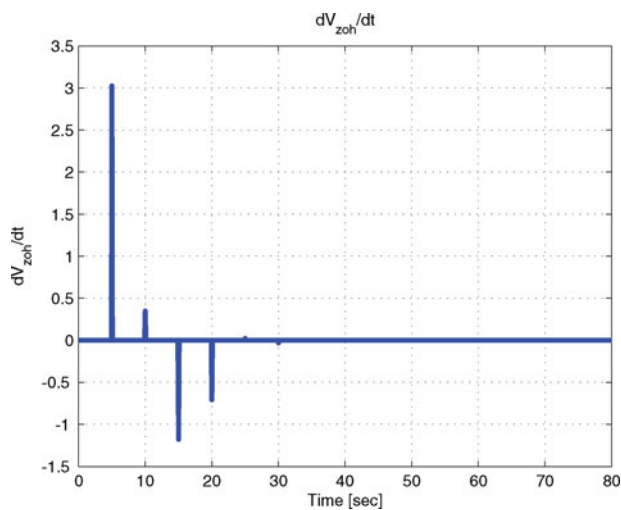
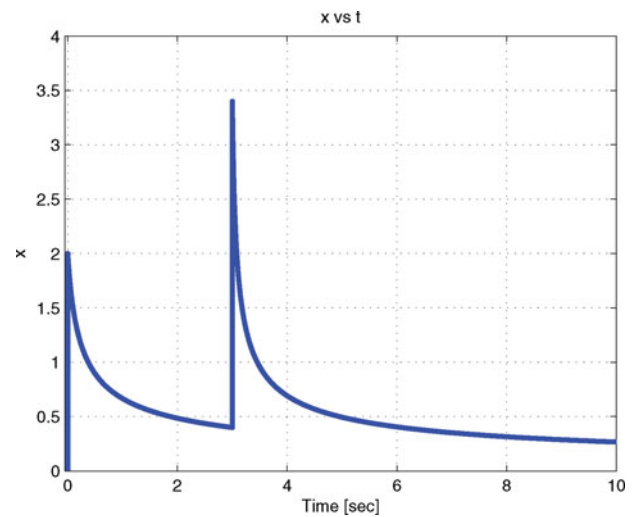


Figure 6. Simple example – tracking error after ZOH.

Figure 7. Simple example – V_{Zoh} .Figure 8. Simple example – $\dot{V}_{Zoh}$.Figure 9. Integrating with δ functions.

of this paper is only stability analysis, we will only mention that the following developments cover all those systems that *have* solutions. Nevertheless, the illustration of discontinuous and yet stable systems does not necessarily imply that the mathematical analysis can take care of such cases and it is the scope of this paper to show that appropriate mathematical tools do exist or can be developed.

It is useful to mention that LaSalle's original works on stability theory and in particular his Invariance Principle, that extended Lyapunov stability theory to include the cases when the derivative was only negative *semi-definite*, were initially dealing only with autonomous systems of the form $\dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x}(t))$ and thus the Lyapunov function $V(\mathbf{x}(t))$ and its derivative $\dot{V}(\mathbf{x}(t))$ were *not* explicit functions of the time t . As we already mentioned, when LaSalle's following works were extending the Invariance Principle to

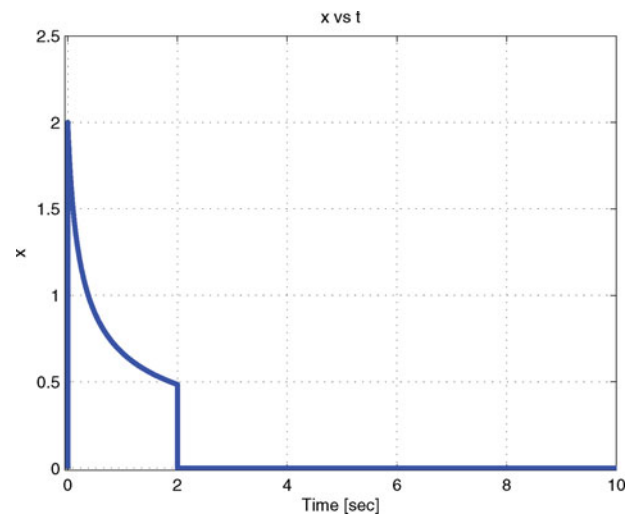


Figure 10. Reaching equilibrium point in finite time.

nonautonomous systems of the form $\dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x}(t), t)$, even if the Lyapunov function remains $V(\mathbf{x}(t))$, the derivative $\dot{V}(\mathbf{x}(t), t) = \frac{\partial V(\mathbf{x})}{\partial \mathbf{x}} \frac{d\mathbf{x}}{dt} = \frac{\partial V(\mathbf{x})}{\partial \mathbf{x}} \mathbf{f}(\mathbf{x}(t), t)$ can still be an explicit function of t . Because, as many publications claim, ‘at times can be difficult to determine even the definiteness of such functions,’ LaSalle’s original works only dealt with those Lyapunov derivatives that satisfy a relation of the form $\dot{V}(\mathbf{x}, t) \leq -W(\mathbf{x}) \leq 0$.

However, as we subsequently show, although in many cases this relation could be sufficient, it may still *unnecessarily* restrict the applicability of stability theory. For example, while for $\dot{V}_1(x, t) = -x^2(2 + \sin t)$ one can define $W(x) = x^2$ and then write $\dot{V}_1(x, t) \leq -W(x) \leq 0$, this is not possible for $\dot{V}_2(x, t) = -x^2(1 + \sin t)$. Nevertheless, one can still define $W(x) = x^2$ and $g(t) = 1 + \sin t$ and get $\dot{V}_2(x, t) \leq -W(x)g(t) \leq 0$. It is clear that $\dot{V}_2(x, t) \leq 0$ or, in other words, that $\dot{V}_2(x, t)$ is *uniformly* negative semi-definite. In a more general case, such as $\dot{V}_3(x, t) = -x_1^2(1 + \sin t) - x_2^2(1 + \cos t)$ it is still clear that $\dot{V}_3(x, t)$ is *uniformly* negative semi-definite although it cannot be written in any one of the (more convenient) forms. Therefore, whenever needed, we will show that one can directly deal with *explicit* functions of time if they are *uniformly* positive or negative semi-definite.

Under the new formulation, satisfaction of one of the following two assumptions must be checked *along the trajectories of the system* under investigation:

(A) $|\mathbf{f}(\mathbf{x}(t), t)|$ is bounded for any *bounded* $\mathbf{x}$.

or

(B) $|\int_{\alpha}^{\beta} \mathbf{f}(\mathbf{x}(\tau), \tau) d\tau|$ is bounded for any *finite* time interval $\gamma = \beta - \alpha$.

Note: The following development will show how helpful the satisfaction of *any one* of these two assumptions is for the stability analysis. Nevertheless, the reader may wonder, and rightfully so, why these assumptions would actually hold in a general nonlinear systems. Indeed, at this stage these are only *assumptions* that must be *checked*, because they may or may not necessarily hold. However, we will show below that they do hold in most cases of interest. Furthermore, it is important to emphasise that, instead of dealing with properties of the general function $\mathbf{f}(\mathbf{x}, t)$, the Invariance Principle approach only tests its properties ‘along the trajectories of the system.’ We again emphasise that it is easy to see that $f(x, t) = x + \sin t$ satisfies assumption A ‘along bounded trajectories,’ while $f(x, t) = x/(1 - x) + \sin t$ or $f(x, t) = x + e^t$ do not. On the other hand, depending on the specific system and on its trajectories, all three examples may or may not satisfy assumption B. If assumption A is satisfied, it actually implies assumption B, yet as shown below, assumption B is satisfied in most situations, even when the velocity $\mathbf{f}(\mathbf{x}, t)$ is not bounded and, instead, may contain impulses, diverging exponentials, etc. Finally, it is useful to note that, as far as stability is concerned, $\mathbf{f}(\mathbf{x}, t)$ and its integral are *not required to be continuous* and that stability theory based on the Invariance Principle

approach only needs the guarantee that $\mathbf{x}(t)$ *cannot* pass an *infinite* distance in *finite* time. At first glance, assumption B could seem to imply that $\mathbf{f}(\mathbf{x}, t)$ is required to be a $L_1[\alpha, \beta]$ function. However, similarly to assumption A, assumption B is also a property of the function $\mathbf{f}(\mathbf{x}, t)$ only along the system *trajectories* and therefore it may or may not hold for same function. Also, the trajectory $\mathbf{x}(t)$ does not have to be continuous in order to satisfy assumption B, yet it is useful to emphasise that it can only contain a finite number of discontinuities along any finite length of time.

Theorem 2.3 (The new Invariance Principle): *Consider the nonlinear nonautonomous system (1). Assume that there exists a positive definite Lyapunov function $V(x)$ and that its derivative $\dot{V}(x, t)$ along the trajectories of Equation (1) is negative semi-definite, i.e., satisfies $\dot{V}(x, t) \leq 0$. For any initial condition $x(t = 0) = x_0$ define the domain $\Omega_0 = \{x | V(x) \leq V(x_0)\}$. Because the Lyapunov derivative is negative semi-definite, it is clear that all system trajectories are bounded and contained within the domain Ω_0 . Now, define the domains $\Omega_e = \{x | \dot{V}(x, t) = 0\}$ and $\Omega_i = \{x | \dot{V}(x, t) \equiv 0\}$ (i.e. in Ω_i the Lyapunov derivative is identically zero, namely, not just equal to zero). Then, if one of the assumptions A or B holds, all limit points of the bounded trajectory $x(t)$ belong to the domain $\Omega_f = \Omega_0 \cap \Omega_e$. In particular, limit points that are also equilibrium points of $x(t)$ belong to $\Omega_f = \Omega_0 \cap \Omega_i$. (See Appendix for the complete proof.)*

Note: In the new Invariance Principle, we differentiate between different kinds of limit points. First, we have isolated limit points of the rosette type, that the trajectories may reach, then leave and come back to them an infinite number of times. Then, we may have equilibrium points that the trajectory approaches in order to ultimately stay there. In the rosette case, the Lyapunov derivative is zero at the limit point, yet its derivatives are not zero, so next moment the derivative itself is not zero and the trajectory moves away. On the other hand, a trajectory that approaches an equilibrium point comes there to stay, so all derivatives of the Lyapunov derivative are also zero. As we already hinted and as following examples will illustrate, in nonautonomous systems we must also differentiate between what we call ‘regular’ equilibrium points that (similarly to autonomous systems) once the trajectory starts there or reaches them at any time, it stays there, and ‘ultimate’ equilibrium points that the trajectory reaches and remains there only as time tends to infinity. Similarly, we have limit cycles that, once the trajectory reaches, it remains there thereafter, so because it implies that all points that the trajectory passes are defined by $\Omega_e = \{x | \dot{V}(x, t) = 0\}$, it again implies that the Lyapunov derivative remains zero, or in other words, the limit cycles are also defined by $\Omega_i = \{x | \dot{V}(x, t) \equiv 0\}$. We emphasise that, while using the definition $\Omega_i = \{x | \dot{V}(x, t) \equiv 0\}$ is very efficient in locating equilibrium points and limit cycles, the use of $\Omega_e = \{x | \dot{V}(x, t) = 0\}$ may allow finding limit points

of type rosette. However, as examples will illustrate, the usefulness of the equality relation is very doubtful, because without supplementary knowledge about a particular system, it is hard to differentiate between these limit points of type rosette and all other points where at one time the Lyapunov derivative just occasionally happens to be zero, with no special meaning whatsoever.

Note: As we noted above, the readers may have difficulties accepting assumption A and, in particular, assumption B. Therefore, although the assumptions were presented first, so that they can be mentioned in the theorem, in order to facilitate their understanding, it is better to follow the theorem in the following order: In the first part of the theorem, the positive definite Lyapunov function and its negative semi-definite derivative guarantee that all trajectories are bounded. Because $\mathbf{x}(t)$ is thus bounded and because in most cases $\mathbf{f}(\mathbf{x}, t)$ contains 'decent' functions of $\mathbf{x}$ and t , in general it is easy to see whether $\mathbf{f}(\mathbf{x}, t)$, which is the velocity $\dot{\mathbf{x}}(t)$, is bounded. However, as we show below, sometimes $\mathbf{f}(\mathbf{x}, t)$ may contain unbounded functions, such as impulses, diverging exponentials, etc., so boundedness of $\mathbf{f}(\mathbf{x}, t)$ cannot be guaranteed. Nevertheless, this is not a problem that may affect stability, because even in this case, the first part of Theorem 2.3 still guarantees that the state $\mathbf{x}(t)$ is bounded. As a result, the *integral* of $\mathbf{f}(\mathbf{x}, t)$, which is a measure of a segment of a bounded trajectory that the system passes in a finite amount of time, *cannot* be unbounded *unless* the trajectory contains an infinite number of jumps (discontinuities) within any finite length of time. Therefore, although assumption B seems to be more difficult to accept, actually is the assumption that holds in most cases.

Note also that this work emphasises the *identity* relation instead of simple *equality* relation $\Omega = \{\mathbf{x} | \dot{V}(\mathbf{x}, t) = 0\}$. Examples below will show how the identity relation sharpens the stability conclusions in general and, in particular, how it may help reaching the desired asymptotic stability conclusion whenever the case allows. Nevertheless, as one reviewer observed, sometimes this definition of Ω requires special attention. For example, for such a scalar system as $\dot{x} = -e^{-t}x$ (that we discussed in the introductory part to this section) one can select $V(x) = x^2$ and get the derivative $\dot{V}(x, t) = -2x^2e^{-t}$ and it is easy to conclude that $x = 0$ is the only 'limit set.' However, this is only apparent, as one can easily verify that no trajectory that starts at $x \neq 0$ reaches $x = 0$. As this example and various examples will also show (see, in particular, Example 2.6 below), for such limit sets as 'regular' equilibrium points the identity $\dot{V}(\mathbf{x}, t) \equiv 0$ must hold for *all* t . Nevertheless, for such limit sets that are only 'ultimate' equilibrium points, the same identity that defines the domain $\Omega = \{\mathbf{x} | \dot{V}(\mathbf{x}, t) \equiv 0\}$ must only be satisfied as $t \rightarrow \infty$. In other words, once the trajectory starts at an equilibrium point it remains there, while more general domains of attraction are reached by the trajectory *only* as time tends to infinity. For the specific example above, while $\mathbf{x} = 0$ is

a 'regular' equilibrium point as it satisfies $\dot{V}(\mathbf{x}, t) \equiv 0$ for all t , *all other finite* values of $\mathbf{x}$ satisfy the identity as time tends to infinity. Therefore, while no nontrivial trajectory reaches $\mathbf{x} = 0$, this is still an equilibrium point for the trivial 'trajectory' $\mathbf{x}(t) = 0$. On the other hand, while any other point is an eventual point of attraction for some trajectory, it is only an 'ultimate' equilibrium point, as no trajectory starting at finite time at $\mathbf{x} \neq 0$ remains there ever after.

Nevertheless, for such an example as $\dot{x} = -(1 + \sin t)x$, the same $V(\mathbf{x}) = x^2$ results in the derivative $\dot{V}(\mathbf{x}, t) = -2x^2(1 + \sin t)$. Here, $x = 0$ is the only equilibrium point and indeed it guarantees $\dot{V}(\mathbf{x}, t) \equiv 0$ for all t including $t \rightarrow \infty$.

Therefore, the following sections and examples will attempt to show that proper use of the Invariance Principle ultimately leads to proper detection and location of the various limit sets of the particular system under analysis. More on the practicality and usefulness of defining the limit set by the *identity* relation instead of simple *equality* will be added after first example below.

Also, note that, while assumption A is identical to assumption 1 in LaSalle's formulation, our presentation of assumption B above is different from LaSalle's original presentations. In LaSalle's formulation, assumption 2 is written in the form $\int_{\alpha}^{\beta} |\mathbf{f}(\mathbf{x}(\tau), \tau)| d\tau = \mu(\beta - \alpha)$. The function $\mu(\tau)$ is a 'modulus of continuity,' from which one can imply that the trajectory is supposed to be a continuous function of time. Here, we replaced the modulus of continuity by a simple bound, because although continuity is desirable, cannot be guaranteed in practical environments and it is not necessarily needed for stability. For another simple illustration, one may compare the 'decent' uniformly continuous converging function $x_1(t) = e^{-t}$ with the equally converging, though not continuous, ladder function described by $x_2(t) = e^{-(k+1)}$ for $k < t \leq k+1$, $k=0, 1, 2, \dots$. Nor is continuity needed for stability under the Invariance Principle approach, because, as the proof of stability shows, the only condition that is needed is that the trajectory, continuous or not continuous, *cannot* pass an infinite distance in finite time.

Finally, previous formulations restricted the discussion to those systems where one could write $\dot{V}(\mathbf{x}, t) \leq -W(\mathbf{x}) \leq 0$. In this case, the limit points belong to the special domain Ω that could be defined as $\Omega = \{\mathbf{x} | W(\mathbf{x}) \equiv 0\}$. However, the examples above (and many applications below) show that in many practical situation one cannot define Ω as above, yet nonetheless Ω can be defined as $\Omega = \{\mathbf{x} | W(\mathbf{x})g(t) \equiv 0\}$, $\Omega = \{\mathbf{x} | \sum W_i(\mathbf{x})g_i(t) \equiv 0\}$, or just $\Omega = \{\mathbf{x} | \dot{V}(\mathbf{x}, t) \equiv 0\}$, all implying that the limit sets are those particular points that can make $\dot{V}(\mathbf{x}, t)$ to be *identically* zero as time tends to infinity (or *uniformly* identically zero for equilibrium points). As we prove in this paper, any one of the definitions above is legitimate definition that is covered by the new Invariance Principle.

Furthermore, even for the basic case, when it is possible to write $\dot{V}(\mathbf{x}, t) \leq -W(\mathbf{x}) \leq 0$, one must be careful. Although it is true that any limit point must belong to $\Omega = \{\mathbf{x} | W(\mathbf{x}) = 0\}$ it does not imply that the inverse is true. In other words, except for simple cases, many if not most points in this Ω are not limit points. Take for example $\dot{V}(x, t) = \dot{V}(x_1, x_2, x_3, \dots, x_n, t) = -x_1^2(1 + \sin t) - x_2^2 \leq -x_2^2 \leq 0$. One selects $W(x) = x_2^2$ and reaches the conclusion that the limit set is $x_2 = 0$. However, if one does not forget that $\dot{V}(x, t) = -x_1^2(1 + \sin t) - x_2^2 \leq 0$, it is quite clear that any $x_1 \neq 0$ makes $\dot{V}(x, t) \leq 0$ and so, cannot be a limit point. On the other hand, points defined by $x_1 = 0$ and $x_2 = 0$ do belong to the limit set. This example may suggest that direct use of the Lyapunov derivative is preferable in all cases.

In other words, under fairly mild conditions, the Invariance Principle extension to nonautonomous systems guarantees that all trajectories ultimately reach that part of the domain Ω that is bounded by the initial conditions. However, as examples below illustrate, its significance and efficiency and sometimes even its mere existence seems to have remained unknown to a large section of potential users.

Example 2.4: The following example from Slotine and Li (1991) represents the closed-loop error dynamics of an adaptive control system for a first-order plant with one unknown parameter and it was used in Hou, Duan, and Guo (2010) for an illustration of Barbalat's Lemma application to stability analysis

$$\begin{aligned}\dot{e}(t) &= -e(t) + \theta w(t) \\ \dot{\theta}(t) &= -e(t)w(t).\end{aligned}\quad (6)$$

The input function $w(t)$ makes the system nonautonomous. Selecting the Lyapunov function $V(t) = e^2(t) + \theta^2(t)$ results in the derivative $\dot{V}(t) = -2e^2(t) \leq 0$. Because, as stated in Slotine and Li (1991), 'one cannot conclude the convergence of $e(t)$ because the dynamics is nonautonomous,' Slotine and Li (1991) impose conditions on $w(t)$ that allow computing the second derivative $\ddot{V}(\mathbf{x}, t)$ in order to reach the conclusion that $\dot{V}(\mathbf{x}, t)$ is uniformly continuous. In this case, application of Barbalat's Lemma indicates that $e \rightarrow 0$ as $t \rightarrow \infty$. Because the conditions for stability are restrictive, the same example was then again used in a recent publication (Hou et al., 2010) as an illustration for their own attempted mitigation of Barbalat's Lemma conditions. After checking such 'mitigated' conditions as 'only boundedness of $\dot{e}(t)$ and thus uniform continuity of $e(t)$,' the authors feel entitled to apply their own version of Barbalat's Lemma and reach the conclusion that $e \rightarrow 0$ as $t \rightarrow \infty$. Still, there is no clear conclusion that can be drawn with respect to θ . This result is more restricted than it may look at first sight, because it does not say whether the set includes all points on the $e(t) = 0$ axis, just some points that the trajectories pass when they occasionally cross the axis, or if ultimately the entire motion occurs only along this

axis. Besides, one may ask what could happen if the control function $w(t)$ is such that does not necessarily guarantee uniform continuity.

Instead, a full response is ultimately provided by the Invariance Principle extension to nonautonomous systems. First, because the system satisfies (the fairly mild) assumption A for any bounded $w(t)$, LaSalle's Invariance Principle directly tells us that the state-vector $\{e, \theta\}$ ends within the domain defined by $W(e(t)) = 2e^2(t) = 0$ which immediately provides the same result $e(t) = 0$ as above, without having to worry about continuity of $\dot{V}(\mathbf{x}, t)$. Furthermore, because the new Invariance Principle emphasises the condition $W(e(t)) \equiv 0$ and this in turn implies that the derivatives of $W(e(t))$ are also zero at $e(t) = 0$, we differentiate the function $W(e(t))$. The first derivative does not add much, because $\dot{W}(e(t)) = -4e(t)\dot{e}(t) = 0$ at $e(t) = 0$ for any value of $\dot{e}(t)$. However, the second derivative $\ddot{W}(e(t)) = -4(\dot{e}(t))^2 - 4e(t)\ddot{e}(t) = -4(\dot{e}(t))^2 = 0$ implies $\dot{e}(t) = 0$. Substituting in both equations also leads to $\theta = 0$ and $\dot{\theta}(t) = 0$. In other words, the fairly straightforward result of the new Invariance Principle for this nonautonomous system is *asymptotic stability*.

Note: This example is a good illustration for the implications of our emphasis on the identity relation $W(\mathbf{x}(t)) \equiv 0$, instead of simply writing $W(\mathbf{x}(t)) = 0$. Note that for any fixed point $\mathbf{p}$ in plane, $W(\mathbf{p})$ is constant, and so is for all those particular points $\mathbf{p}$ where $W(\mathbf{p}) = 0$. However, for the state vector $\mathbf{x}(t)$ moving along some trajectory, $W(\mathbf{p}) = 0$ means $W(\mathbf{p}) = \lim \{W(\mathbf{x}(t)) \rightarrow \mathbf{p}\} = 0$. Therefore, if the derivative $\dot{W}(\mathbf{p}) = \lim (dW(\mathbf{x}(t)) \rightarrow \mathbf{p})/dt \neq 0$, this implies that next moment $W(\mathbf{x}(t)) < 0$ and so, $W(\mathbf{x}(t))$ has moved away from $\mathbf{p}$ and therefore $\mathbf{p}$ cannot be a limit point. Thus, while all limit points do belong to the set defined by $W(\mathbf{p}) = 0$, the converse is not true, namely, not all, and actually very few, points of the set defined by $W(\mathbf{p}) = 0$ are limit points. Still, because for a general function the identity relation $W(\mathbf{p}) \equiv 0$ seems to imply that $W(\mathbf{x}(t))$ must be infinitely differentiable, on the face of it the condition may seem to be very restrictive or even impossible to satisfy. However, we recall that a system of order n is fully defined by n derivatives or, in state-space representation, by the first derivative of all state-variables. In a similar way to all conclusions of Lyapunov-like techniques, this condition also depends on the proper selection of the Lyapunov function and sometimes one is confined to only partial results. Still, as we only require locating a function $W(\mathbf{x}(t))$ that satisfies $\dot{V}(\mathbf{x}, t) \leq -W(\mathbf{x}(t)) \leq 0$, if a first selection $W_1(\mathbf{x}(t))$ is not satisfactory, one may continue looking for $W_2(\mathbf{x}(t))$, $W_3(\mathbf{x}(t))$, etc., to find a more appropriate function. These are general remarks because, as the example above (as well as the many examples to follow) illustrates, using the definition $W(\mathbf{p}) \equiv 0$ managed to reduce the large set defined by $e(t) = 0$ to only the origin $\{e(t) = 0, \theta(t) = 0\}$ and to asymptotic stability, as illustrated in Figure 11, where $w(t)$ is a square-wave input command.

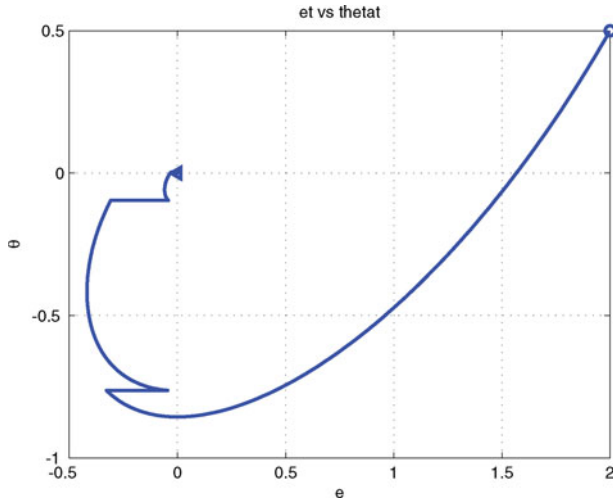


Figure 11. Example 2.4 – typical trajectory with square-wave $s_{1,2,3}$.

Furthermore, assume that the input command $w(t)$ contains one or more impulse functions. In this case, no continuity can be guaranteed and so, all previous conditions, including the original LaSalle's assumptions, are violated. Nevertheless, as peculiar as it may appear at first look, the negative semi-definite derivative still implies that all trajectories are bounded and thus the new assumption B holds and therefore, even in this case the *asymptotic stability* of the system is preserved.

Example 2.5: The advantage of using the Invariance Principle with nonautonomous nonlinear systems is even more evident in those cases when the Lyapunov derivative may take more complex forms and the assumably necessary conditions of continuity and differentiability are obviously violated. For example, define the square function wave $s_{\alpha,\beta,T}$ with lower value α , upper value β and period T and assume that the system is

$$\begin{aligned}\dot{e}(t) &= -e(t)s_{1,2,3} + \theta w(t) \\ \dot{\theta}(t) &= -e(t)w(t).\end{aligned}\quad (7)$$

The same Lyapunov function $V(e, \theta) = e^2(t) + \theta^2(t)$ results in the derivative $\dot{V}(e, \theta, t) = -2e^2(t)s_{1,2,3} \leq 0$. Here, although the input command $w(t)$ does not show in the Lyapunov derivative, the square-wave $s_{1,2,3}$ does and therefore the derivative function is neither continuous in t nor differentiable, so Barbalat's Lemma cannot help any longer and (as may usually happen in adaptive control) the square-wave (or any other nondifferentiable function, no matter how small) may look like a dangerous, modelled or unmodelled, dynamics that may even threaten the stability of the system, unless special correction devices are used. Instead, because $\dot{V}(e, \theta, t) = -2e^2(t)s_{1,2,3} \leq -2e^2(t) \leq 0$, LaSalle's Invariance Principle avoids the continuity requirements and ends with the conclusion $e(t) \rightarrow 0$ as $t \rightarrow \infty$.

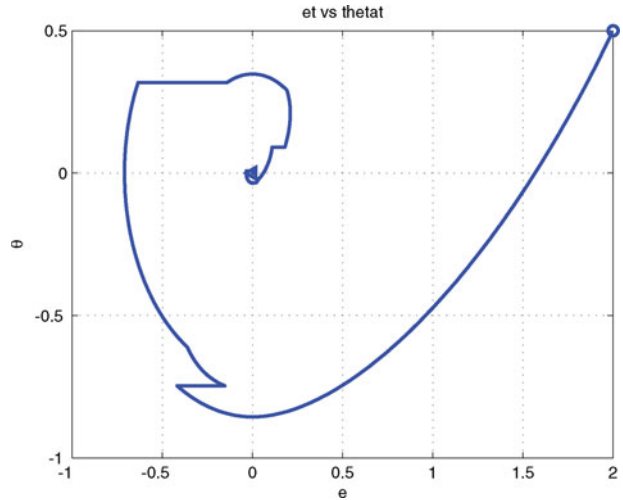


Figure 12. Example 2.5 – typical trajectory with square-wave $s_{0,2,3}$.

Finally, the new Invariance Principle avoids all those apparent problems and just leads to exactly the same conclusion of asymptotic stability.

Furthermore, if we now replace the previous square-wave by $s_{0,2,3}$ (with lower value 0, upper value 2 and period 3), we no longer can find a function $W(e(t))$ to write $\dot{V}(e, \theta, t) = -2e^2(t)s_{0,2,3} \leq -W(e(t)) \leq 0$ and the Invariance Principle in its original formulation cannot be applied either. However, the new Invariance Principle and the direct use of the nonautonomous Lyapunov derivative helps, because we still get $\dot{V}(e, \theta, t) = -2e^2(t)s_{0,2,3} \leq 0$, so $e(t) \equiv 0$ guarantees that $\dot{V}(e, \theta, t) \equiv 0$ for all times and this still leads to asymptotic stability, as illustrated in Figure 12.

Example 2.6: Based on a similar claim that 'the Invariance Principle cannot be directly applied to nonlinear nonautonomous systems,' another publication (Lee, Liaw, & Chen, 2001) uses the following 'counterexample' that is meant to illustrate the limits of applicability of LaSalle's theory in nonautonomous systems:

$$\begin{aligned}\dot{x}_1(t) &= e^{-2t}x_2(t) \\ \dot{x}_2(t) &= -e^{-2t}x_1(t) - x_2(t).\end{aligned}\quad (8)$$

For stability analysis, the authors chose the simple quadratic Lyapunov function $V(\mathbf{x}) = \frac{1}{2}(x_1^2 + x_2^2)$. It is easy to see that the Lyapunov derivative is $\dot{V}(\mathbf{x}) = -x_2^2$. Then, because, as the analysis of Lee et al. (2001) claims, the origin $\{x_1 = 0, x_2 = 0\}$ is, apparently, the only equilibrium point and because the system satisfies assumption A, the natural conclusion should be asymptotic stability. Therefore, because the paper shows that for any initial conditions $\{x_1(0) \neq 0, x_2(0) = 0\}$ the system ends with some other $x_1 \neq 0$ and $x_2 = 0$, this seems to show that the direct application of LaSalle's Invariance Principle to nonautonomous systems is problematic.

Indeed, because at first look the analysis of Lee et al. (2001) seems to be correct, one would be naturally led to look for possible weaknesses in the proofs of the Invariance Principle. However, because the proof is also correct, a second look at system (8) is required and indeed it shows that the time-varying terms make the analysis of this system more complex than what the first analysis might have revealed. Although, as claimed in Lee et al. (2001), the origin $\{x_1 = 0, x_2 = 0\}$ is an equilibrium point, it is by no means the *only* equilibrium point for this system. This little observation actually provides another good motivation for emphasising *identical* vs. equal in Theorem 2.3 above and the note afterward that the identity must hold for all time only for ‘regular’ equilibrium points, while for ‘ultimate’ equilibrium points, it must only hold as time tends to infinity. Because the conclusion of the Invariance Principle is that the system ends within the domain in state space defined by $x_2 \equiv 0$ that implies $x_2 = 0$ and $\dot{x}_2 = 0$, etc., substituting both relations into the second equation leaves only $e^{-2t}x_1(t) = 0$. Again, while $x_1(t) = 0$ is the solution for any *finite* time t , the situation is different when time tends to infinity, when $\lim_{t \rightarrow \infty} (e^{-2t}x_1(t)) = 0$ is satisfied for *any* finite value of x_1 . Therefore, any point on the entire x_1 axis (limited by $V(\mathbf{x}) \leq V(\mathbf{x}_0) = x_{1_0}^2/2$ or $|x_1| \leq |x_{1_0}|$) can be an ultimate equilibrium point for some trajectory of this system. It is also worth noticing that, although no nontrivial trajectory ends at $\{x_1 = 0, x_2 = 0\}$, the origin is the equilibrium point for the ‘trajectory’ $\{x_1(t) = 0, x_2(t) = 0\}$ that starts, stays and ends at the origin. Thus, the so-called ‘counterexample’ actually is an excellent illustration that not only is the Invariance Principle fit for stability analysis of nonlinear nonautonomous systems, but that it also provides the proper answer for more general situations, such as those systems that are not necessarily asymptotically stable (see Figure 13 for various initial conditions) and again forces us to differentiate between limit sets that are also

‘regular’ equilibrium points and those limit sets that are ‘ultimate’ equilibrium points. Thus, the conclusion of Lee et al. (2001) is based on a misinterpretation of the result and the example above actually is a perfect illustration of the Invariance Principle *efficiency*.

Example 2.7: Another example that does not satisfy previously apparently necessary conditions is:

$$\begin{aligned}\dot{x}_1(t) &= (1 + \sin t)x_2(t) \\ \dot{x}_2(t) &= -(1 + \sin t)x_1(t) - (1 + \cos t)x_2(t).\end{aligned}\quad (9)$$

For stability analysis, we again chose the simple quadratic Lyapunov function $V(\mathbf{x}) = \frac{1}{2}(x_1^2 + x_2^2)$. It is easy to see that the Lyapunov derivative is $\dot{V}(\mathbf{x}) = -(1 + \cos t)x_2^2$. Although there is no function $W(\mathbf{x})$ to write $\dot{V}(\mathbf{x}) \leq -W(\mathbf{x}) \leq 0$, it is clear that $\dot{V}(\mathbf{x}) = -(1 + \cos t)x_2^2 \leq 0$ and that $\dot{V}(\mathbf{x}) \equiv 0$ implies $x_2 \equiv 0$, which in turn gives $x_2 = 0$ and $\dot{x}_2 = 0$ and, by substituting in the equations, $x_1 = 0$ and $\dot{x}_1 = 0$. Here, the origin $\{x_1 = 0, x_2 = 0\}$ is the only equilibrium point and because the system satisfies assumption A, the conclusion is asymptotic stability (see Figure 14).

Example 2.8: The example above illustrates the difficulties that could be encountered in stability analysis of complex systems. We emphasise that the Invariance Principle is only interested in the limit points, or in other words in those solutions of $W(\mathbf{x}(t)) = 0$ that exist for $t \rightarrow \infty$. However, on one hand the equation may also supply many other solutions that are not relevant to the stability issue, and on the other hand, as the example above specifically illustrates, one may have difficulties identifying exactly those solutions of interest that exist only for $t \rightarrow \infty$. An even more interesting and more common example is the linear nonautonomous

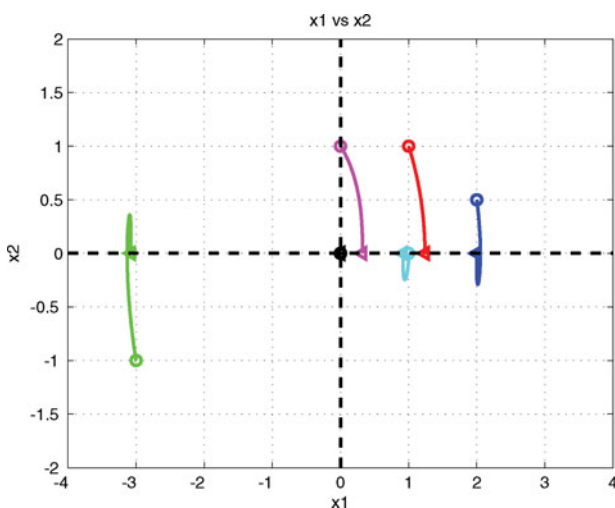


Figure 13. Example 2.6 – typical trajectories.

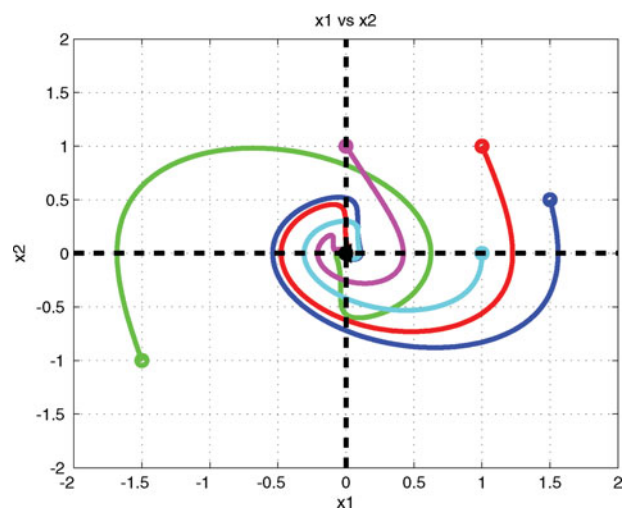


Figure 14. Example 2.7 – typical trajectories.

damped oscillator:

$$\ddot{q}(t) + a(t)\dot{q}(t) + q(t) = 0. \quad (10)$$

Here, $a(t)$ is some time-varying damping factor that makes the system nonautonomous. As long as things are continuous, both Barbalat's Lemma and the Invariance Principle can be applied. We use this example because, similarly with the previous one, one must carefully interpret the results in order to reach the right conclusion. Define $q = x_1$, $\dot{q} = x_2$, to get

$$\begin{aligned} \dot{x}_1(t) &= x_2(t) \\ \dot{x}_2(t) &= -a(t)x_2(t) - x_1(t). \end{aligned} \quad (11)$$

Choose the Lyapunov function $V(\mathbf{x}) = \frac{1}{2}|x|^2$ to get the derivative 'along the trajectories' $\dot{V}(\mathbf{x}, t) = -a(t)x_2^2$. As long as $a(t)$ is continuous in time and $\mathbf{x}(t)$ is continuous, Barbalat's Lemma can be applied and, at least, tells us that ultimately, $x_2 \rightarrow 0$ as $t \rightarrow \infty$. We now apply the Invariance Principle, because of the fine mathematical points that must be clarified. First of all as long as $a(t)$ is bounded, the system satisfies assumption A and the Invariance Principle can be applied. For convenience and without loss of generality, assume $a(t) > 1$ and because $\dot{V}(\mathbf{x}, t) = -a(t)x_2^2 \leq -x_2^2$, the Invariance Principle allows us to know that the system ends with $x_2(t) \equiv 0$. Because in turn this implies $\dot{x}_2(t) = 0$, $\ddot{x}_2(t) = 0$, etc., we substitute in the first equation to get $\dot{x}_1(t) = 0$ and from the second equation $x_1(t) = 0$. Therefore, for a bounded damping function, the system demonstrates asymptotic stability (see Figure 15).

Example 2.9: However, one gets an even more complex and interesting case if $a(t)$ is not bounded as, for example, if the damping is $a(t) = 2 + e^t$. For the

same Lyapunov function as above, we get the same Lyapunov derivative $\dot{V}(\mathbf{x}, t) = -a(t)x_2^2 = -(2 + e^t)x_2^2 \leq -x_2^2 \leq 0$ and from here, that $\mathbf{x}(t)$ is bounded. However, because of the unbounded $a(t)$, this time it is hard to call $\dot{V}(\mathbf{x}, t)$ 'uniformly continuous' and possible use of Barbalat's Lemma seems to be blocked. Nevertheless, before we conclude anything else using the Invariance Principle, we must check the assumptions. Because of the unbounded $a(t)$, assumption A is not satisfied either. However, because we know that all trajectories are bounded, assumption B is satisfied and therefore, we can go on with the Invariance Principle analysis. Because again we get $x_2(t) \equiv 0$, we again end with $x_2(t) = 0$ and $\dot{x}_2(t) = 0$. From here, it would be easy to conclude, like in previous case, that $\{0, 0\}$ is the equilibrium point and that the system should demonstrate asymptotic stability. However, because of the unbounded term in the second equation, this time one must be more careful, because there could be more than one limiting point as time tends to infinity. So, as one analyses the second equation, one must take special care of the term $(2 + e^t)x_2(t)$. Here, one cannot just substitute $x_2 = 0$ and cancel the term because on nontrivial trajectories, $x_2(t)$ is not necessarily zero at finite time. Even as time tends to infinity, it is not sufficient to assume that the value $x_2 \rightarrow 0$ cancels the unbounded term factor, because if, for example, $x_2(t)$ tends to zero as $\frac{1}{t}$, the product would still tend to infinity. However, this is also not possible, because we know that all trajectories are bounded. Apparently, here the Invariance Principle can only tell that ultimately $x_2(t)$ ends with $x_2 = 0$, while $x_1(t)$ is just bounded. However, we may also observe that, in order to end with a bounded value of the term $(2 + e^t)x_2(t)$, $x_2(t)$ must tend to zero as some exponential function. Trying a solution of the form $\dot{q}(t) = x_2(t) = k_1 e^{-\alpha t}$ and then substituting it along with $q(t) = x_1(t) = k_2 - k_1 e^{-\alpha t}/\alpha$ and $\ddot{q}(t) = -\alpha k_1 e^{-\alpha t}$ into the system equation and after some algebra, results in

$$(\alpha - 1)k_1 e^{-\alpha t} - \alpha k_1 e^{(1-\alpha)t} - \alpha k_2 = 0. \quad (12)$$

Because this is supposed to be a relation among constant coefficients, Equation (12) requires $\alpha = 1$ and then $k_2 - k_1 = 0$. Therefore, the solution is $q(t) = x_1(t) = k_1(1 - e^{-t})$, $\dot{q}(t) = x_2(t) = k_1 e^{-t}$ for any coefficient k_1 . Thus, while all trajectories end with $x_2 = 0$, the other co-ordinate ends at $x_1 = k_1$, such that the limit point for each trajectory is $\{x_1 = k_1, x_2 = 0\}$. It is important to notice that, although the limit set is the horizontal axis (bounded by the initial conditions) that could lead to an eventual limit cycle, in this case, each trajectory asymptotically reaches a well-defined limit point (see Figure 16). In the examples above, the point $\{x_1 = 0, x_2 = 0\}$ appears as a member of the limit set, because the origin is the 'limit point' for the trivial 'trajectory' $\{x_1(t) = 0, x_2(t) = 0\}$.

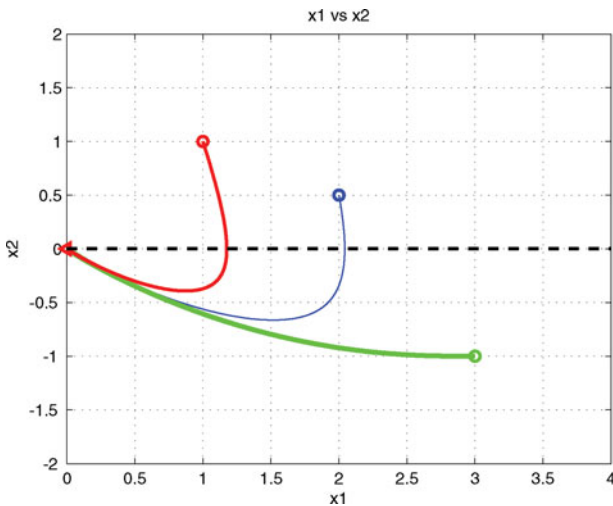


Figure 15. Example 2.8 – typical trajectories with bounded damping.

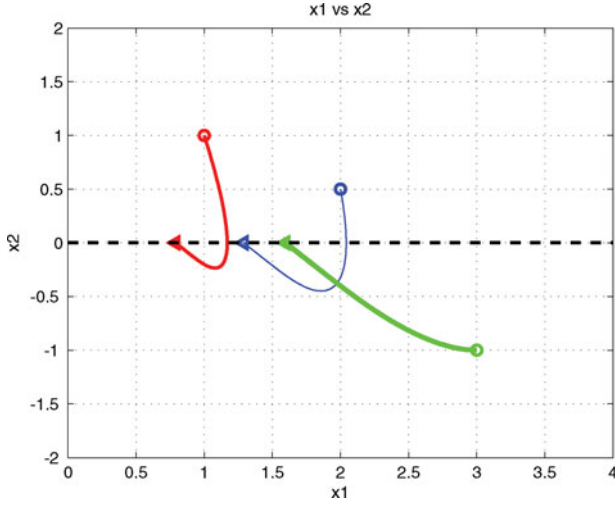


Figure 16. Example 2.9 – typical trajectories with unbounded damping.

2.5 The limiting equation

Returning to the previous example with transient terms, one notes that in order to ‘overcome’ an apparent problem with the Invariance Principle, the authors of Lee et al. (2001) then mention Artstein’s concept of limit equation (Artstein, 1976, 1977, 1978), and use it to develop their own extension that is apparently needed to deal with the example above. We recall that the limit equation deals with systems that can be separated into two parts as shown below.

Theorem 2.10 (The Limiting Equation Theorem): Consider the nonlinear nonautonomous system (1). Assume that $\mathbf{f}(\mathbf{x}, t) = \mathbf{g}(\mathbf{x}, t) + \mathbf{h}(\mathbf{x}, t)$ where $\mathbf{h}(\mathbf{x}, t)$ is a transient term defined as $\lim_{t \rightarrow \infty} (\mathbf{h}(\mathbf{x}, t)) = 0$. Now, consider the limiting equation

$$\dot{\mathbf{x}}(t) = \mathbf{g}(\mathbf{x}, t) \quad (13)$$

that satisfies one of the assumptions of Theorem 2.2. If the resulting limiting set Ω for Equation (13) is bounded and is asymptotically approached by the trajectories of Equation (13), then Ω is also asymptotically approached by all bounded solutions of Equation (1) (Artstein, 1976, 1977, 1978).

2.6 The Modified Invariance Principle

Although in some cases it may facilitate reaching the same conclusion we reached above about the meaning of $\dot{V} \equiv 0$, the limiting equation idea is more powerful and so, separation of the system into ‘good’ and ‘not good’ parts does not add much as long as the Lyapunov derivative is negative semi-definite. As we show below, it could be more useful to combine Artstein idea and the original Invariance Principle and express it as a Modified Invariance Principle (in terms of Lyapunov function and Lyapunov derivative). This

would be an extension of the regular Invariance Principle to those cases where, due to transient terms, the Lyapunov derivative cannot be called even semi-definite and so, would include the regular Invariance Principle as a particular case.

Theorem 2.11 (The Modified Invariance Principle):

Consider the nonlinear nonautonomous system (1). Let $V(\mathbf{x})$ be a differentiable function bounded from below. (Note that $V(\mathbf{x})$ is not required to be positive definite.) Assume that the derivative $\dot{V}(\mathbf{x}, t)$ along the trajectories of Equation (1) has the form $\dot{V}(\mathbf{x}, t) = W_1(\mathbf{x}, t) + W_2(\mathbf{x}, t)$ such that $W_1(\mathbf{x}, t) \leq 0$ and where $W_2(\mathbf{x}, t)$ is continuous in $\mathbf{x}$ and piece-wise continuous in t and $\lim_{t \rightarrow \infty} (W_2(\mathbf{x}, t)) = 0$. Here, we define a new domain, Ω_M , that contains all bounded trajectories of the system. Under the same conditions and definitions of the regular Invariance Principle, if any one of the assumptions A or B holds, then all bounded trajectories of Equation (1) ultimately reach the domain $\Omega_f = \Omega \cap \Omega_M$ (see Barkana, 1983; Kaufman et al., 1998 for first formulations and Appendix A for the complete proof of the theorem).

Note that the definition of the domain Ω_M is needed for the proofs of stability because it implies that $V(\mathbf{x})$ must remain bounded along any trajectory in this domain. In other words, there exists some M such that $V(\mathbf{x}(t)) = V(\mathbf{x}_0) + \int_0^t \dot{V}(\mathbf{x}(\tau), \tau) d\tau \leq M < \infty$ for any t (including $t \rightarrow \infty$) and along any trajectory in Ω_M . This is a very general result and allows analysis of complex nonlinear systems even when one cannot find the best Lyapunov function. In particular cases, one must analyse the meaning of set Ω that could turn out to be a collection of equilibrium points, limit cycles, etc., yet in many cases it could also be possible to show that all trajectories are bounded and that the entire domain Ω_f is actually reduced to a unique equilibrium point.

Thus, the regular Invariance Principle becomes a corollary of the Modified version for the case $W_2(\mathbf{x}, t) = 0$. For the general case, one way towards defining the meaning of bounded trajectories is to observe that the system might diverge only along those trajectories where the transient term makes $\dot{V}(\mathbf{x}, t)$ positive. One way of dealing with this issue is trying to rewrite $\dot{V}(\mathbf{x}, t)$ in the form $\dot{V}(\mathbf{x}, t) \leq k(t)V(\mathbf{x})$. In this form, the Lyapunov function formulation of the Modified Invariance Principle allows using the Gronwall–Bellman Lemma.

Lemma 2. (Gronwall–Bellman Lemma): If $\dot{V}(t) \leq k(t)V(t)$, where $V(t) \geq 0$, $k(t) \geq 0$, then $V(t) \leq V(0) \exp \left[\int_0^t k(\tau) d\tau \right]$.

Example 2.12: Because the regular Invariance Principle is sufficient for Example 2.6 above, we add an additional transient term that could have complicated the analysis:

$$\begin{aligned} \dot{x}_1(t) &= e^{-2t} x_2(t) \\ \dot{x}_2(t) &= -e^{-2t} x_1(t) - x_2(t) + e^{-t}. \end{aligned} \quad (14)$$

For the same Lyapunov function as before, one gets the derivative $\dot{V}(\mathbf{x}) = -x_2^2 + e^{-t}x_2(t)$. Here, because of the transient term, one cannot even claim that $\dot{V}(\mathbf{x})$ is semi-definite and therefore some trajectories might diverge. Thus, both the Limiting Equation theorem and the Modified Invariance Principle may look modest, as they only claim that all *bounded* solutions finally end within the domain defined by the semi-definite part of the derivative, or in other words by the same $x_2 \equiv 0$. This does not seem to add too much for control systems, where the eventual existence of unbounded trajectories is *the* problem. However, in this case it is again the role of the analysis to find out if there are trajectories that could be unbounded. To this end, we only check those situations where $\dot{V}(\mathbf{x})$ could be positive and this may occur only when $x_2(t) > 0$. If the systems tend to diverge through large values of $x_2(t)$, the negative quadratic term makes $\dot{V}(\mathbf{x})$ negative again. However, divergence may also occur through $x_1(t)$, even if $x_2(t)$ remains small. For all these situations we write

$$\dot{V}(\mathbf{x}) = -x_2^2 + e^{-t}x_2(t) \leq e^{-t}x_2(t). \quad (15)$$

For large $x_1(t)$ or $x_2(t)$ one gets

$$\dot{V}(\mathbf{x}) \leq e^{-t}x_2^2(t) \leq e^{-t}(x_1^2(t) + x_2^2(t)) \quad (16)$$

or

$$\dot{V}(\mathbf{x}) \leq 2e^{-t}V(\mathbf{x}). \quad (17)$$

According to Gronwall–Bellman Lemma

$$V(t) \leq V(0)e^{2\int_0^t e^{-\tau} d\tau} \leq V(0)e^2. \quad (18)$$

Therefore, *all* trajectories are bounded and even in this case, the Modified Invariance Principle gives the adequate response to the problem at task (see Figure 17).

Example 2.13: Again, it is useful to emphasise the advantage of using this paper formulation of the Invariance Principle with nonlinear and nonautonomous systems in those cases when the Lyapunov derivative may take more complex forms. For example, we again use the ‘problematic’ square-wave defined above and assume that the system is

$$\begin{aligned} \dot{x}_1(t) &= e^{-2t}x_2(t) \\ \dot{x}_2(t) &= -e^{-2t}x_1(t) - x_2(t)s_{0,2,5} + e^{-t}. \end{aligned} \quad (19)$$

For the same Lyapunov function as before, one gets the derivative $\dot{V}_1(x) = -x_2^2(t)s_{0,2,5}$ for the first system (without transient term) or $\dot{V}_2(x) = -x_2^2s_{0,2,5} + e^{-t}x_2(t)$ for the system with additional transient term. In both cases the Lyapunov derivative is not continuous, so the alternative of Barbalat’s Lemma cannot be used and apparently no

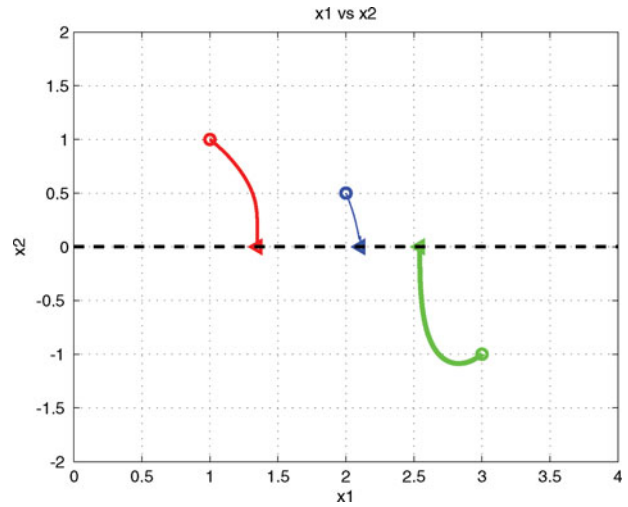


Figure 17. Example 2.12 – typical trajectories.

conclusion is to be drawn. Furthermore, because of the square-wave factor, there is no nontrivial function $W(x)$ that would allow writing $\dot{V}_1(x) = -x_2^2(t)s_{0,2,5} \leq -W(x) \leq 0$, so the original Invariance Principle apparently is also not of use. However, because by defining $W(x) = x^2$ one gets $\dot{V}_1(x) = -x_2^2(t)s_{0,2,5} \leq -W(x)s_{0,2,5} \leq 0$, in both cases the new Invariance Principle simply ends with same conclusion as above, namely $x_2(t)$ vanishes as time tends to infinity and $x_1(t)$ ends with some ultimately constant value (see Figure 18).

While global asymptotic stability could be desirable, trying to force asymptotic stability conclusions may overlook or even totally miss other important results. Moreover, the above results are hardly disappointing. If $x_2(t)$ is ‘the plant’ while $x_1(t)$ is the nonlinear or adaptive ‘gain,’ the conclusion of the Invariance Principle in both cases is that the ‘plant’ demonstrates asymptotic stability, while the ‘gain’

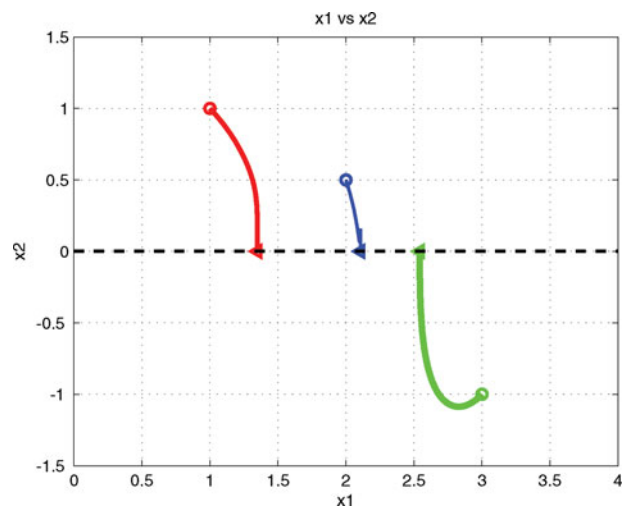


Figure 18. Example 2.13 – typical trajectories.

ends with the *guarantee* of ultimately constant values, the desired aim of any adaptive, and probably any nonlinear, controller.

In a general Adaptive Control environment, even when the adaptation ends with perfect tracking, apparently, the adaptive gains may not reach any limit and may even continue wandering for ever, unless the presence of some ‘persistent excitation’ is guaranteed. Such situations and more complex real-world examples, where the combination of Invariance Principle, Gronwall–Bellman Lemma and other mathematical tools had led to the proof of ultimate adaptive gain convergence in general MIMO Adaptive Control systems, can be found in Barkana (2005, 2008).

3. Beyond global asymptotic stability

Interestingly enough, although this paper was written in order to raise awareness towards LaSalle’s ideas and the various versions of the Invariance Principle, both Barbalat’s Lemma and the Invariance Principle could be further extended and then used in more general situations, when, otherwise, all that is known is that all trajectories, or only *some* trajectories, are bounded.

Example 3.1: For a first illustration, we consider the system

$$\begin{aligned}\dot{e}(t) &= -e(t) + e^3(t) + \theta w(t) \\ \dot{\theta}(t) &= -e(t)w(t).\end{aligned}\quad (20)$$

We chose a complex example and it proved to be more complex than initially intended. Selecting the Lyapunov function $V(e, \theta) = (e^2(t) + \theta^2(t))/2$ results in the derivative $\dot{V}(e, \theta) = -e^2(t) + e^4(t)$. One can see that the Lyapunov derivative is negative semi-definite only for those $e(t)$ within the domain defined by $|e| < 1$. At first look, one may instinctively decide that trajectories that start within the domain defined by $|e| < 1$ are bounded. However, the analysis of this case is more complex than a first look might indicate. For this specific system, the fact that the Lyapunov derivative is negative semi-definite in some region only implies that the Lyapunov function, a norm of the state-vector $\{e, \theta\}$, starts decreasing within the domain defined by $|e| < 1$. However, it does not imply that $e(t)$ remains within the domain defined by $|e| < 1$. Only initial values $\{e_0, \theta_0\}$ that are within the domain defined by $e^2(t) + \theta^2(t) < 1$ do *fully* guarantee that the trajectories remain within the domain defined by $|e| < 1$ and are, therefore, bounded. However, this is the general problem of Lyapunov stability approach that requires the Lyapunov derivative to *always* be negative definite (or, for us, at least negative semi-definite) and so, it is not a result of the present analysis. Similarly, only initial values $\{e_0, \theta_0\}$ that are within the domain defined by $e^2(t) + \theta^2(t) > 1$ and $|e| > 1$ ‘guarantee’ that the Lyapunov derivative will always

remain positive and will lead to instability. In between, there is a region where the trajectories could be either stable or unstable. One may decide to only use the guaranteed domain of stability or to keep looking for better Lyapunov functions. In any case, we want to prove that, even if the derivative of Lyapunov function is not definite along *all* trajectories of the system, all *bounded* trajectories nonetheless end within the domain Ω defined by $\Omega = \{e, \theta | \dot{V}(e, \theta) \equiv 0\}$. In our case, $\dot{V}(e, \theta) = -e^2 + e^4 = e^2(e^2 - 1) = 0$ is satisfied at $e = 0$, $e = +1$, or $e = -1$. Any trajectory that starts within the domain defined by $e^2(t) + \theta^2(t) < 1$ cannot reach $|e| = 1$ and therefore, will ultimately end with $e \equiv 0$. In this case, when the limit ‘set’ of a trajectory is reduced to a single point, the simpler assumptions of the Invariance Principle are sufficient, without requiring the more demanding uniform continuity condition. Again, because this implies $e = 0$ and $\dot{e} = 0$ and assuming $w(t) \neq 0$, these trajectories also end with $\theta = 0$. Therefore, it is clear that trajectories that start within $e^2(t) + \theta^2(t) < 1$ remain bounded and ultimately reach the origin. Moreover, those trajectories that start within $|e| < 1$ yet outside the unit circle also remain bounded and reach the origin if the norm decreases below 1 *before* they leave the domain $|e| < 1$. However, other trajectories that start within $|e| < 1$ yet outside the unit circle diverge if the norm remains above 1 and they leave the domain $|e| < 1$ (see Figure 19). Still, other bounded trajectories that start outside this circle may end along one of the lines defined by $e = -1$ or $e = 1$.

Example 3.2: For a different situation, we assume that the system is

$$\begin{aligned}\dot{e}(t) &= e(t) - e^3(t) + \theta(t)w(t) \\ \dot{\theta}(t) &= -e(t)w(t).\end{aligned}\quad (21)$$

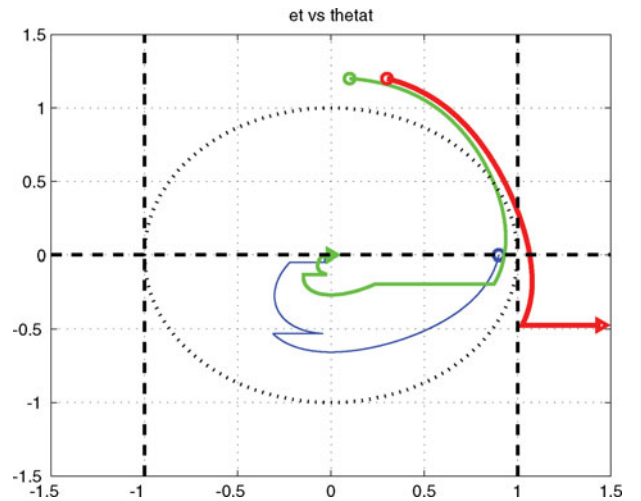


Figure 19. Example 3.1.

This ‘small’ change actually illustrates (and is meant to illustrate) how difficult stability analysis can be for any one of (or all of) Lyapunov–LaSalle–Barbalat approaches. Selecting the same Lyapunov function $V(e, \theta) = (e^2(t) + \theta^2(t))/2$ results in the derivative $\dot{V}(e, \theta) = e^2(t) - e^4(t)$. In this case, one can see that the Lyapunov derivative is negative semi-definite for $e(t)$ within the domain defined by $|e| \geq 1$ and positive semi-definite for those $e(t)$ within the domain defined by $|e| < 1$. Here, the only clear and direct result is that any trajectory that starts within the circle defined by $e^2(t) + \theta^2(t) < 1$ cannot stay there as the norm *must* increase. However, the trajectory may still be bounded because, outside this circle, *most* of the remaining domain will be dominated by $\dot{V}(e, \theta) \leq 0$. Therefore, many trajectories will be bounded, yet not necessarily all. Also, $\dot{V}(e, \theta) = e^2(t) - e^4(t) = e^2(1 - e^2) = 0$ is satisfied at $e = 0$, $e = +1$ or $e = -1$. Because any trajectory that starts within the domain defined by $|e| < 1$ cannot reach $|e| = 0$, it is very attractive to conclude that all bounded trajectories must ultimately end along one of the lines defined by $e \equiv 1$ or $e \equiv -1$. However, because each one of the lines separates the domains $\dot{V}(e, \theta) < 0$ and $\dot{V}(e, \theta) > 0$, one cannot eliminate the possibility that the bounded trajectories may for ever oscillate between the two domains. Here, in principle, Barbalat’s Lemma could help, because although not sufficiently emphasised, its rather tough uniform continuity requirement replaces and obviates the requirement $\dot{V}(e, \theta) \leq 0$, as long as one knows that $V(e, \theta)$ is bounded and ‘has a finite limit.’ However, even for bounded trajectories, it is not clear that $V(e, \theta)$ reaches any limit and also there is no reason to assume that uniform continuity of $\dot{V}(e, \theta)$ is satisfied. Thus, if there is no supplementary information, the partial conclusions above are all that stability analysis can provide in this case. Indeed, although simulations seem to indicate that various bounded trajectories do ultimately reach same limit cycle, they also show *discontinuities* and thus, the limit cycle keeps moving from the domain $\dot{V}(e, \theta) < 0$ to $\dot{V}(e, \theta) > 0$ and vice-versa and is *not* defined by $\dot{V}(e, \theta) \equiv 0$ (see trajectories in Figure 20, the Lyapunov function in Figure 21 and Lyapunov derivative in Figure 22 with square-wave input command).

Similar situations also hold when the Lyapunov derivative $\dot{V}(\mathbf{x}, t)$ is an explicit function of time. For these domains where one clearly has $\dot{V}(\mathbf{x}, t) \leq -W(\mathbf{x}) \leq 0$ or $\dot{V}(\mathbf{x}, t) \leq -W(\mathbf{x})g(t) \leq 0$, the Invariance Principle assumptions are sufficient to guarantee that all trajectories of this domain are bounded and ultimately reach the domain defined by $W(\mathbf{x}) \equiv 0$.

3.1 A Generalised Invariance Principle

Therefore, carefully, we can state a Generalised Invariance Principle.

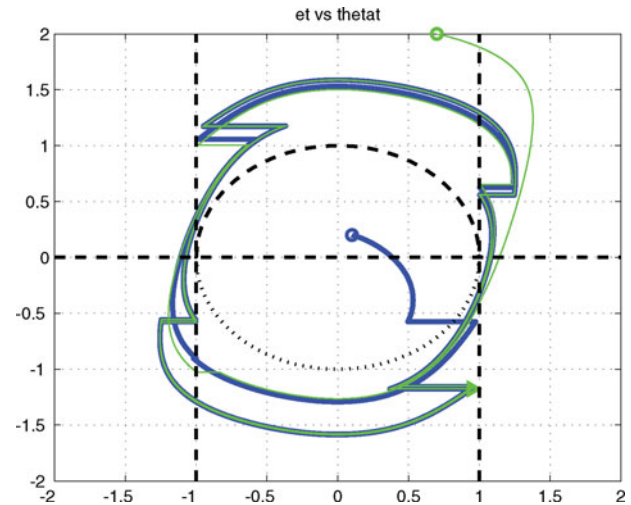


Figure 20. Example 3.2: trajectories.

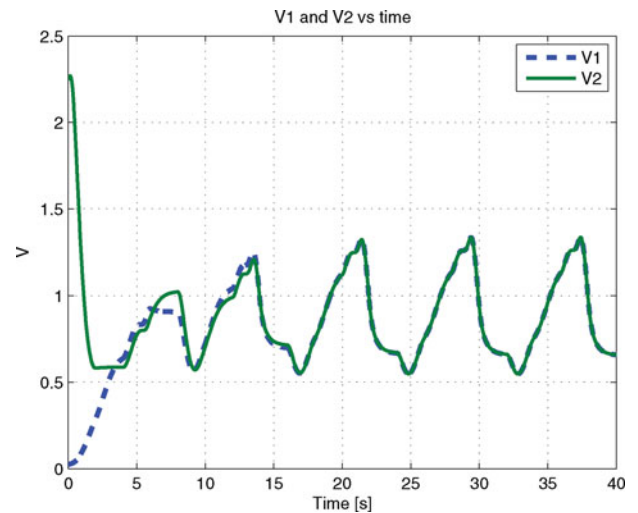


Figure 21. Example 3.2: Lyapunov function.

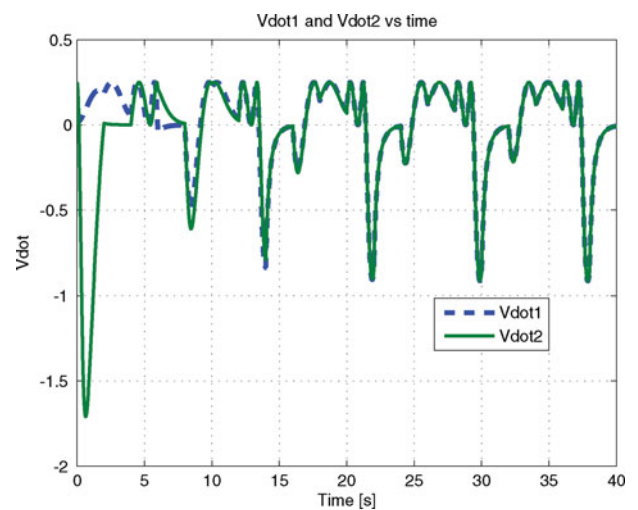


Figure 22. Example 3.2: Lyapunov derivative.

Theorem 3.3 (A Generalised Invariance Principle):

Consider the nonlinear nonautonomous system (1). Let $V(\mathbf{x})$ be a differentiable function bounded from below and let $\dot{V}(\mathbf{x}, t)$ be its derivative along the trajectories of Equation (1). For any closed and bounded domain around the origin where one can write $\dot{V}(\mathbf{x}, t) \leq 0$, if any one of the assumptions A or B holds, then all trajectories of Equation (1) are bounded and ultimately reach the domain $\Omega_i = \{x | W(x) \equiv 0\}$. If, however, $\dot{V}(\mathbf{x}, t) \geq 0$ holds within a closed and bounded domain around the origin and $\dot{V}(\mathbf{x}, t) \leq 0$ holds outside, then all trajectories are globally bounded, yet none reaches the origin. In general, they may oscillate from one domain to another. If one can show that $V(\mathbf{x})$ ultimately ends with finite value, uniform continuity of $\dot{V}(\mathbf{x}, t)$ may still guarantee that bounded trajectories reach the domain $\Omega_e = \{x | \dot{V}(\mathbf{x}, t) = 0\}$ (Appendix A).

Note the emphasis on *identical* that, if satisfied, can lead to sharper conclusion than simply *equal*. As the following example shows, in some conditions, $\dot{V}(\mathbf{x}, t) = 0$ implies $\dot{V}(\mathbf{x}, t) \equiv 0$. In such a case, once the trajectory reaches the domain $\Omega_e = \{x | \dot{V}(\mathbf{x}, t) = 0\}$, it stays there. However, in some situations, when same trajectory can interchangeably move between $\dot{V}(\mathbf{x}, t) < 0$ and $\dot{V}(\mathbf{x}, t) > 0$, most probably the Invariance Principle result is not applicable and then, uniform continuity could be useful to at least know that ultimately, the bounded trajectories end within the domain defined by $\dot{V}(\mathbf{x}, t) = 0$.

Example 3.4: For another illustration, we change the example above to be

$$\begin{aligned}\dot{e}(t) &= e(t) - e^3(t) - e(t)\theta^2(t) + \theta(t)w(t) \\ \dot{\theta}(t) &= \theta(t) - \theta^3(t) - e^2(t)\theta(t) - e(t)w(t).\end{aligned}\quad (22)$$

Selecting the same Lyapunov function $V(e, \theta) = (e^2(t) + \theta^2(t))/2$ results in the derivative

$$\dot{V}(e, \theta) = e^2(t) + \theta^2(t) - (e^4(t) + \theta^4(t) + 2e^2(t)\theta^2(t))$$

$$\dot{V}(e, \theta) = e^2(t) + \theta^2(t) - (e^2(t) + \theta^2(t))^2$$

$$\dot{V}(e, \theta) = (e^2(t) + \theta^2(t))(1 - (e^2(t) + \theta^2(t))). \quad (23)$$

One can see that the Lyapunov derivative is positive definite for $e^2(t) + \theta^2(t) < 1$ and negative semi-definite for $e^2(t) + \theta^2(t) \geq 1$. In other words, we know that all trajectories are bounded, yet, as mentioned, this first result by itself does not necessarily eliminate the possibility of everlasting oscillations between the domains defined by $\dot{V}(e, \theta) < 0$ and $\dot{V}(e, \theta) > 0$. However, if uniform continuity of $\dot{V}(e, \theta)$ is satisfied, in this case all trajectories end with $(e^2(t) + \theta^2(t))(1 - (e^2(t) + \theta^2(t))) = 0$ that is satisfied at the origin, where $e^2(t) + \theta^2(t) = 0$, or at $e^2(t) + \theta^2(t) = 1$. The origin is an unstable equilibrium point, as any trajectory that starts within the domain defined by $e^2(t) + \theta^2(t) < 1$

cannot reach the origin and therefore all trajectories should ultimately end along the circle defined by $e^2(t) + \theta^2(t) = 1$. If this result is not certain because we must show that $V(e, \theta)$ has a finite limit, stability theory cannot and does not supply all results. To test whether Barbalat's expected result holds here, we can re-write Equation (23) as

$$\dot{V}(t) = 2V(t)(1 - 2V(t)). \quad (24)$$

One can see that $\dot{V}(t) = 0$ if $V(t) = \frac{1}{2}$. Further differentiations lead to $\ddot{V}(t) = \dot{V}(t)P_1(V), \dots, V^{(k)}(t) = \dot{V}(t)P_{k-1}(V), \dots$, where the various $P_i(V)$ are polynomials in V . Therefore, the factor $\dot{V}(t) = 0$ makes all derivatives zero at $V(t) = \frac{1}{2}$ which indeed implies that ultimately, $e^2(t) + \theta^2(t) \equiv 1$.

We do not have more information and further analysis of this example is beyond the scope of this paper, yet this result could imply either some limit cycle or just a single limit point. As simulations of this specific example actually show, all limit points are indeed located on the circle $e^2(t) + \theta^2(t) = 1$. In the absence of external input command, trajectories that start outside the circle asymptotically reach single, stable, limit points (Figure 23) on the circle, while the 'trajectory' that starts at a point on the circle remains there for ever. Input commands (in this case, a square-wave) ultimately lead all trajectories to move along the limit cycle (Figure 24).

4. On stability in the sense of Lyapunov

A system is called 'stable in the sense of Lyapunov' if for any ϵ positive and arbitrarily small, there exists some positive δ such that if $|x(0)| \leq \delta$, then $|x(t)| \leq \epsilon$ for any $t \geq 0$. If in addition $x(t) \rightarrow 0$ as $t \rightarrow \infty$, the system is asymptotically stable.

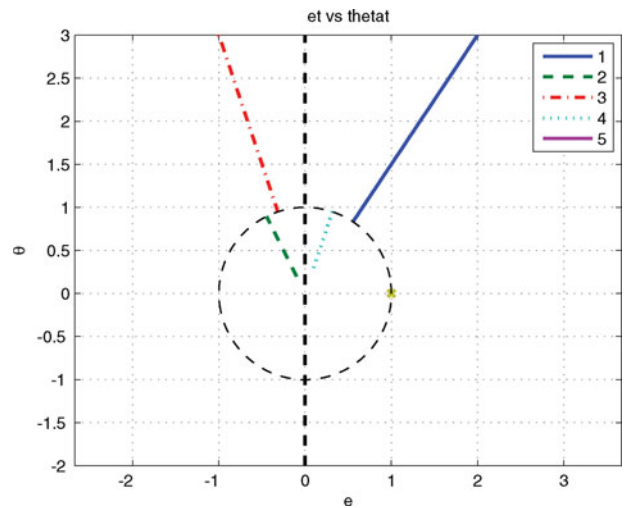


Figure 23. Example 3.4 – without input command.

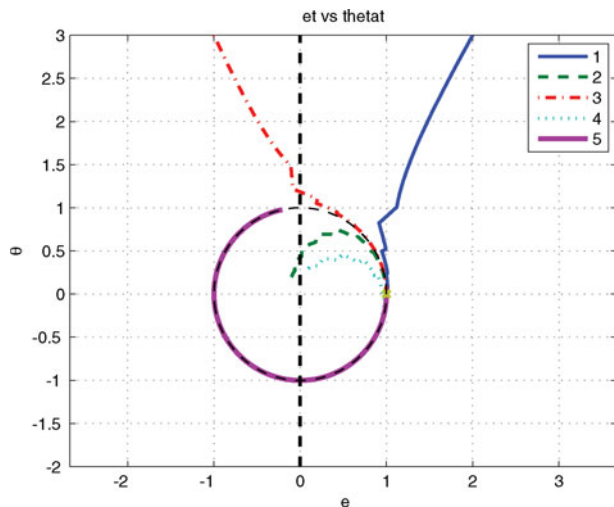


Figure 24. Example 3.4 – with square-wave input command.

Therefore, one may ask whether the system can be called stable or asymptotically stable ‘in the sense of Lyapunov’ only because all trajectories are bounded and even if they ultimately converge to the origin.

The answer depends on the kind of Lyapunov function one manages to fit to a specific system. In the simple case of Example 1 above, where $V(t) = e^2(t) + \theta^2(t)$, the negative semi-definite derivative $\dot{V}(t)$ guarantees that for any $\epsilon = V_0$, trajectories that start within $V(t) \leq \delta = V_0$ will stay within $V(t) \leq \epsilon = V_0$. In the more general case, if the Lyapunov function $V(\mathbf{x})$ is positive definite and the derivative is negative semi-definite and therefore $V(\mathbf{x})$ is bounded by the initial value $V(0)$, one can easily relate it to the norm of $\mathbf{x}(t)$ through relations of the form $a|\mathbf{x}(t)| \leq V(\mathbf{x}) \leq b|\mathbf{x}(t)|$ and it is not difficult to relate the limits to ϵ and δ .

In the most general case, however, when the Lyapunov function $V(\mathbf{x})$ is only required to be bounded from below, even if its derivative $\dot{V}(\mathbf{x}, t)$ is negative semi-definite, stability is defined by the guarantee that all trajectories are bounded and asymptotic stability if they all eventually tend to zero, yet not in the classical sense of Lyapunov.

5. Conclusion and further work

By showing the usefulness and efficiency of LaSalle’s Invariance Principle, this paper is a late and modest homage to LaSalle’s original contribution to stability analysis of nonautonomous nonlinear systems. Furthermore, although the main aim of this paper is an attempt to raise awareness to LaSalle’s contribution to stability theory, it is believed that the new Invariance Principle versions presented in this paper go beyond any results so far. Besides, it should not be read as critical to Barbalat’s result. First of all, as already

mentioned, Barbalat’s Lemma was very useful before any extensions of LaSalle’s ideas to nonautonomous systems were available. It is also believed that only the strict and narrow interpretation of uniform continuity is a problem, because it might seem to imply that any step, pulse or impulse function may destroy the proofs of stability by Barbalat. It is beyond the scope of this paper, yet merely *ultimate* uniform continuity could be sufficient. In other words, if there is some finite time t_1 such that the function $f(t) = \dot{V}(\mathbf{x}, t)$ is uniformly continuous only for the time $t \geq t_1$, this should not affect Barbalat’s result. Furthermore, when it is known that $\dot{V}(\mathbf{x}, t) \leq 0$, piece-wise continuity should also suffice. (Note: to avoid ‘counterexamples’ originating in such functions as $(\sin t)^t$, maybe we should call ‘piece-wise *uniformly* continuous’ those functions that are *uniformly* continuous along *all* intervals.) However, because in these cases the new versions of Invariance Principle free the analysis from almost any requirement of continuity, it is not clear what could even milder interpretations of Barbalat’s Lemma add. Last but not least, after writing so much in order to defend LaSalle’s contribution, it also appears that, at least for the uniform continuity case, the interpretation of Barbalat in the context of stability was very restricted and unjust. Although the time-function $h(t)$ may reach zero many times along its time-history, if we prove that it approaches zero as $t \rightarrow \infty$, this actually may imply that it goes to settle there, or in other words, to loose any dynamics. Here we note that it is easy to think of functions that can be counterexamples to this claim. However, in the context of *system stability*, points that do not guarantee that the Lyapunov derivative $f(t) = \dot{V}(\mathbf{x}, t)$ tends to be *identically* zero as time tends to infinity are not limit point of the trajectory. Considering examples such as e^{-t} , $t^n e^{-t}$, $\frac{1}{t}$, $\frac{\sin t}{t}$, etc., one sees that all end with no dynamics at infinity, so (even if one finds some appropriate counterexamples) at least *in the context of stability analysis*, maybe Barbalat’s Lemma should be read ‘the uniformly continuous $h(t)$ tends to be identically zero as $t \rightarrow \infty$,’ or ‘ $h(t)$ tends to zero as $t \rightarrow \infty$ along with its relevant derivatives,’ which, at least for the uniform continuity case, would give much more than what one gets today.

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Appendix A. Proof of Modified (and Generalised) Invariance Principle

As many colleagues commented, the very definition of limit point and limit set can be confusing in nonlinear nonautonomous systems. Therefore, before we proceed with stability analysis of trajectories, it is useful to define limit points (or points of accumulation). Define an infinite sequence of discrete times $t_0, t_1, t_2, \dots, t_i, \dots$ and the corresponding points on the trajectory $\mathbf{x}(t_0), \mathbf{x}(t_1), \mathbf{x}(t_2), \dots, \mathbf{x}(t_i), \dots$. A limit point is defined as such a point about which any *arbitrarily small* neighbourhood contains an infinite number of points $\mathbf{x}(t_i)$ of the trajectory. An example of limit point is a stable equilibrium point that is asymptotically approached by the trajectory as time goes to infinity. Thus, for any neighbourhood, arbitrarily small, around the equilibrium point there exists some finite time t_0 such that *all* points on the corresponding trajectory belong to this neighbourhood for all times larger than t_0 .

Any *bounded* trajectory contains (at least) a limit point. Although this is basically known, we repeat the argument for its usefulness towards the proofs. The discretisation procedure above resulted in an infinite number of points $\mathbf{x}(t_i)$. Because a bounded trajectory is closed within some bounded domain D , all points $\mathbf{x}(t_i)$ are also located within same bounded domain. To prove that at least one point is a limit point according with the definition above, we divide D into two subdomains. It is clear that at least one of them must still contain an infinite number of points. We continue the procedure with that domain. Continuing the same procedure, there will always be (at least) one domain, no matter how small, that still contains an infinite number of points. A very important argument for the proof of stability is that ultimately, we identify (at least) one point of the *bounded trajectory*, that its neighbourhood, no matter how small, still contains an infinite number of points.

There are various ways the limit point may appear when we start moving along the trajectory from $\mathbf{x}(t_0)$ to $\mathbf{x}(t_1), \dots, \mathbf{x}(t_i), \dots$. The simplest form is when the trajectory has an isolated limit point $\mathbf{x}^*$ that it only reaches as time approaches infinity. In such a case, for *any* ε positive and *arbitrarily small* there exists a time t_m such that the distance $|\mathbf{x}(t_i) - \mathbf{x}^*| < \varepsilon$ for all $i > m$.

Another form of isolated limit point is best illustrated by a rosette with all its infinitely many petals having a common point. Although the trajectory may reach the neighbourhood $|\mathbf{x}(t_i) - \mathbf{x}^*| < \varepsilon$ of the limit point $\mathbf{x}^*$ at some (finite or infinite) time t_m , it goes away, yet it comes back to same neighbourhood and then leaves it and comes back an infinite number of times.

The last form of limit point is best illustrated by the trajectory that reaches the neighbourhood $|\mathbf{x}(t_i) - \mathbf{x}^*| < \varepsilon$ of the limit point $\mathbf{x}^*$ at some (finite or infinite) time t_m , then goes away and comes back to same neighbourhood an infinite number of times, yet keeps doing it via similar points, namely, all point it meets on its way are limit points. This is an example of limit cycle, a collection of un-isolated limit points.

For the content of this paper it is important to emphasise that, once the discrete time sequence is defined, whether the original curve was continuous or discontinuous is immaterial if it only was bounded and therefore there exists at least one such point $\mathbf{x}^*$ where the discrete distances $|\mathbf{x}(t_i) - \mathbf{x}^*|$ become smaller and smaller.

After establishing the meaning of limit sets, we try to find methods that can locate the limit sets of system trajectories. Here, it is important to emphasise that, no matter what the results of various approaches may be, the limit points of any trajectories are located *on the trajectory*, even if in order to reach them, we must allow time to actually reach infinity.

It is also important to emphasise that the main argument for all following developments is the fact that $V(\mathbf{p})$ is bounded from below for any point $\mathbf{p}$ along any bounded trajectory. If the point $\mathbf{p}$ is reached at some finite time t , then for any $\mathbf{p}$ there exists a constant M , $0 \leq M < \infty$, such that $V(\mathbf{p}) = V(\mathbf{x}(0)) + \int_0^t \dot{V}(\mathbf{x}(\tau), \tau) d\tau \geq -M > -\infty$ and this remains true if $\mathbf{p}$ is a limit point that is reached as time tends to infinity. In this case, for any *limit point* $\mathbf{p}$ there exists a constant N , $0 \leq N < \infty$, such that $V(\mathbf{p}) = V(\mathbf{x}(0)) + \int_0^\infty \dot{V}(\mathbf{x}(\tau), \tau) d\tau \geq -N > -\infty$.

Now, consider any *bounded* trajectory of Equation (1) and let $\mathbf{p}$ be a limit point of this trajectory. Thus, there exists some finite constant c such that $V(\mathbf{p}) = c < \infty$. We assume $\dot{V}(\mathbf{x}, t) = W_1(\mathbf{x}, t) + W_2(\mathbf{x}, t)$. For convenience, we assume that $W_1(\mathbf{x}, t)$ is uniformly continuous in $\mathbf{x}$ (not in t , though) and $W_2(\mathbf{x}, t)$ is bounded for bounded $\mathbf{x}$ and vanishes as $t \rightarrow \infty$.

Because of the delicate points of the proof, we first restrict the discussion to those situations where one can write $W_1(\mathbf{x}, t) \leq -W(\mathbf{x}) \leq 0$, where $W(\mathbf{x})$ is uniformly continuous and where Ω is then defined as $\Omega = \{\mathbf{x} | W(\mathbf{x}) = 0\}$. It is desired to show that *any* limit point $\mathbf{p}$ must belong to Ω . To show this by a contradiction argument, assume that $\mathbf{p}$ does not belong to Ω . Then, $-W(\mathbf{p}) < 0$. Let $S_\varepsilon(\mathbf{p})$ be a neighbourhood of $\mathbf{p}$, $S_\varepsilon(\mathbf{p}) = \{\mathbf{x} : |\mathbf{x} - \mathbf{p}| \leq \varepsilon\}$, such that, if $\mathbf{x} \in S_\varepsilon(\mathbf{p})$ then $-W(\mathbf{x}) < -d_1$ for some $d_1 > 0$ sufficiently small. Because $W_2(\mathbf{x}, t)$ eventually vanishes, there exists a finite time t_1 such that $\dot{V}(\mathbf{x}, t) \leq -W(\mathbf{x}) + W_2(\mathbf{x}, t) < -d_2 < 0$ for all $t \geq t_1$ and all $\mathbf{x} \in S_\varepsilon(\mathbf{p})$ and for some d_2 , where $0 < d_2 \leq d_1$.

Let us first assume that $\mathbf{x}(t)$ approaches $\mathbf{p}$ asymptotically. In other words, after some finite time, $\mathbf{x}(t)$ enters the neighbourhood of $\mathbf{p}$ and remains in this neighbourhood thereafter. Thus, there exists a time $t_2 \geq t_1$ such that $\mathbf{x} \in S_\varepsilon(\mathbf{p})$ for all $t \geq t_2$. In this case, $V(\mathbf{x}(t)) = V(\mathbf{x}(t_2)) + \int_{t_2}^t \dot{V}(\mathbf{x}(\tau), \tau) d\tau$ and thus $V(\mathbf{x}(t)) \rightarrow -\infty$ as $t \rightarrow \infty$, which contradicts the lower boundedness assumption.

Here, it is important to notice that, although we assumed the continuity of $W(\mathbf{x})$, at no point there was any use or need for the continuity of the trajectory $\mathbf{x}(t)$ itself.

Before one can conclude that *any* limit point $\mathbf{p}$ belongs to Ω , one must eliminate the eventuality of other possible situation that might occur in the case that we called the rosette situation. The limit point could still be located within the domain defined by $-W(\mathbf{x}) < 0$, because the trajectory could also approach the limit point in a different way, namely, $\mathbf{x}(t)$ may enter $S_\varepsilon(\mathbf{p})$ and leave it immediately, and repeat this an infinite number of times. This way, the total amount of time that $\mathbf{x}(t)$ remains within $S_\varepsilon(\mathbf{p})$ could still remain finite and in this case the integral $V(\mathbf{x}(t)) = V(\mathbf{x}(t_2)) + \int_{t_2}^t \dot{V}(\mathbf{x}(\tau), \tau) d\tau$ could remain finite. However, the total distance would still be infinite and by assumptions A or B, the trajectories *cannot* pass an infinite distance in a finite amount of time, and thus the trajectory cannot pass an infinite number of times through point $\mathbf{p}$ within a finite amount of time. One must, therefore, conclude that even in this case $W(\mathbf{p}) = 0$ and therefore $\mathbf{p} \in \Omega$ and, according to the case, $\mathbf{p} \in \Omega_M \cap \Omega$ or $\mathbf{p} \in \Omega \cap \Omega_0$.

Although the proof is ‘almost’ complete, there is another fine point related to the use of *identical* versus *equal*. First, assume that the trajectory does leave the limit point $\mathbf{p}$ and comes back to it an infinite number of times, yet it does it only through other limit points. In this case, once $W(\mathbf{x})$ reaches its zero value at the first limit point, it remains zero thereafter. In other words,

instead of just reaching the domain simply defined by $W(\mathbf{x}) = 0$, the trajectory has reached the domain defined by $W(\mathbf{x}) \equiv 0$.

Limit points of the rosette kind are defined by $\Omega = \{\mathbf{x} | W(\mathbf{x}) = 0\}$ (i.e., simple equality), because the trajectory does leave the point $\mathbf{p}$ that belongs to $\Omega = \{\mathbf{x} | W(\mathbf{x}) = 0\}$, goes through a neighbourhood where $-W(\mathbf{x}) < 0$ and comes back to $\mathbf{p}$ an infinite number of times. However, as many examples show, the domain $\Omega = \{\mathbf{x} | W(\mathbf{x}) = 0\}$ also contains all those points that made $W(\mathbf{x}) = 0$ for just an instant, without having anything to do with limit sets.

Instead, those limit points where the system ultimately comes to settle are characterised by the ultimate loss of dynamics so, similarly to limit cycles, they belong to $\Omega = \{\mathbf{x} | W(\mathbf{x}) \equiv 0\}$.

After one gets used with the proof above, one can follow along the same lines to extend it to the situations where the negative semi-definite Lyapunov derivative *cannot* be written in the convenient form above. As the proof above showed that the transient does not affect the limit points location on *bounded* trajectories, for convenience of the next proof, we ignore the transient terms and move to the next class of systems where $\dot{V}(\mathbf{x}, t)$ can be written as $\dot{V}(\mathbf{x}, t) \leq -W(\mathbf{x})g(t) \leq 0$. In other words, we want to show that the limit points still belong to that domain in plan that guarantee that the Lyapunov derivative is zero. Again, first assume that $\mathbf{p}$ is an equilibrium point that the trajectory approaches asymptotically. Even if $g(t)$ is such that it *occasionally* makes $W(\mathbf{x})g(t)$ zero (such as $V(x, t) = -x^2(1 + \sin t) \leq 0$ or $\dot{V}(x, t) = -x^2 s_{0.1,1} \leq 0$), if $-W(\mathbf{p}) < 0$, then in any neighbourhood of $\mathbf{p}$ that the trajectory enters at time t_0 , one gets $V(\mathbf{x}(t)) = V(\mathbf{x}(t_0)) + \int_{t_0}^t \dot{V}(\mathbf{x}(\tau), \tau) d\tau \rightarrow -\infty$ as $t \rightarrow \infty$. From here, same argument also holds to conclude that $\mathbf{p}$ must belong to $\Omega = \{\mathbf{x} | \dot{V}(\mathbf{x}, t) = 0\}$. Furthermore, if the limit point is an equilibrium point or any limit point where the trajectory ultimately comes to rest, then $\mathbf{p}$ must belong to $\Omega = \{\mathbf{x} | \dot{V}(\mathbf{x}, t) \equiv 0\}$. If the trajectory leaves $\mathbf{p}$ and comes back to it an infinite number of times *via limit points*, it can do it only within the domain that keeps $\dot{V}(\mathbf{x}, t) \leq -W(\mathbf{x})g(t) \equiv 0$. Only points of type rosette are defined by $\Omega = \{\mathbf{x} | \dot{V}(\mathbf{x}, t) = 0\}$ (not *identically* zero), because the trajectory leaves such limit points and comes back to them an infinite number of times via regular trajectory points that are not limit points.

Here, one must still differentiate between ‘ultimate’ equilibrium points that the trajectory only reaches as time tends to infinity and ‘regular’ equilibrium points. As the definition implies and as examples illustrated, if the trajectory starts at a ‘regular’ equilibrium point, it stays there for ever. However, for ‘ultimate’ equilibrium points, if the trajectory starts there or reaches them at finite time, it does not necessarily stay there. Therefore, one can differentiate between regular equilibrium points $\mathbf{p}$ that make $\dot{V}(\mathbf{p}, t) \equiv 0$ for all t and those ‘ultimate’ equilibrium points that only make $\dot{V}(\mathbf{p}, t) \rightarrow 0$ as $t \rightarrow \infty$.

Now, extensions to $\dot{V}(\mathbf{x}, t) \leq \sum W_i(\mathbf{x})g_i(t) \leq 0$ or just $\dot{V}(\mathbf{x}, t) \leq 0$ are straightforward. Following along the lines above, in particular using the integral where the integrand that we wrote is the general $\dot{V}(\mathbf{x}, t)$, all limit points of bounded trajectories in all cases are located within the domain Ω defined as $\Omega = \{\mathbf{x} | \dot{V}(\mathbf{x}, t) \equiv 0\}$.

An important observation is that stability analysis addresses *all* trajectories and does not usually address any trajectory in particular. Therefore, one could be tempted to call Ω ‘the limit set of all trajectories’ and then be surprised that the origin, for example, comes out as a limit point (or by other definitions ‘attractor’), while no actual trajectories actually reach the origin. Instead of reaching the conclusion that the limit point of any bounded trajectory may lie outside the trajectory, one must understand that, instead of being ‘the limit set of all trajectories,’ Ω is the

collection of all limit sets of all trajectories. Examples provided in this paper showed that the origin $\mathbf{x} = 0$ may appear as a solution of $W(\mathbf{x}) = 0$, and therefore as a limit point, even when no trajectory ever reaches the origin, because it still is the 'limit point' of the trivial 'trajectory' $\mathbf{x}(t) = 0$.

For the Generalised Invariance Principle, we start with same idea that *if* there is any bounded trajectory, it contains (at least) a limit point. For those domains or those bounded trajectories where one can write $\dot{V}(\mathbf{x}, t) \leq 0$, *all* trajectories are bounded and the proof continues on same lines as above to show that any limit

point cannot belong to $\dot{V}(\mathbf{x}, t) < 0$ that would bring $V(\mathbf{x}(t))$ to $-\infty$, so it can only belong to the domain Ω defined by $\Omega = \{\mathbf{x} | \dot{V}(\mathbf{x}, t) \equiv 0\}$. For other situations, when $\dot{V}(\mathbf{x}, t)$ could equally be negative definite, positive definite or zero, uniform continuity of $\dot{V}(\mathbf{x}, t)$ along bounded trajectories indicates that *all* bounded trajectories may still end within the domain defined by $\dot{V}(\mathbf{x}, t) = 0$. However, because now $\dot{V}(\mathbf{x}, t)$ is not known to be negative semi-definite, one would need other information to show that $V(\mathbf{x})$ reaches a finite limit.